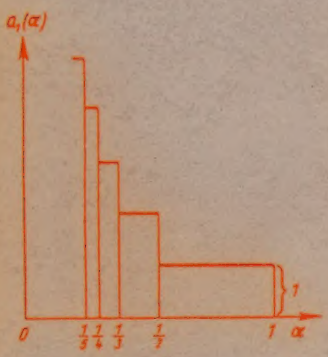
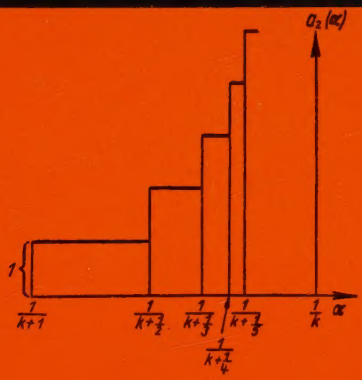
A. Ya. Khintchine





Continued Fractions

Translated by P. Wynn

P. Noordhoff n.v. Groningen

A. Ya. KHINTCHINE:

CONTINUED FRACTIONS

This is a translation of the third and latest edition of the monograph "Continued Fractions" by the late A. Ya. Khintchine. It deals with the properties of the so-called simplest continued fraction

$$a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cdots}}$$

and the uses to which this can be put in the investigation of the arithmetic properties of the continuum, in particular of the laws governing the approximation of algebraic and transcendental numbers by means of rational numbers. The third chapter is devoted to the metric (or probability-theoretic) applications of continued fractions, to which substantial contributions have been made by Soviet mathematicians.
In the Soviet Union, this monograph has run to a number of editions and a series of reprintings.

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CONTINUED FRACTIONS

A. Ya. KHINTCHINE

CONTINUED FRACTIONS

Translated by PETER WYNN



P. NOORDHOFF, Ltd. · GRONINGEN · THE NETHERLANDS

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CONTENTS

Page

Translator's Preface 1
Foreword to the Third Edition 2
Foreword to the Second Edition 3
From the Foreword to the First Edition 3

I. The Properties of the Calculus

§ 1. Introduction 6
§ 2. Convergents 8
§ 3. Infinite Continued Fractions 13
§ 4. Continued Fractions with Natural Elements 17

II. The Representation of Numbers by Means of Continued Fractions

§ 5. The Continued Fraction as an Instrument for Representing Real Numbers 22
§ 6. Convergents as Best Approximations 27
§ 7. The Order of the Approximation 35
§ 8. General Laws of Approximation 41
§ 9. The Approximation of Algebraic Numbers. Liouville's Transcendental Numbers 53
§ 10. Quadratic Irrationals and Periodic Continued Fractions 56

III. The Metric Theory of Continued Fractions

§ 11. Introduction 60
§ 12. The Elements as Functions of the Number Being Represented 61
§ 13. The Metric Approximation of the Rate of Growth of the Elements 69
§ 14. The Metric Estimation of the Rate of Growth of the Denominators of the Convergents. The Main Theorem of the Metric Approximation Theory 74
§ 15. Gauss' Problem and the Theorem of Kusmin 80
§ 16. Mean Values 95

Translator's Preface

The appearance in the English language of the celebrated memoir "Continued Fractions" by the late Alexander Yakovlevitch Khintchine is long overdue. The following is a translation of the third and latest (1961) edition which contains a few footnotes mainly of a bibliographical character and a foreword added by Prof. B. V. Gnedenko. In the translation I have tried to recapture the freshness and vigour of the original. A small number of typographical errors have been corrected.

Mathematical Centre
December 1963

P. Wynn

Foreword to the Third Edition

The present, third edition of the excellent book by A. Ya. Khintchine was undertaken by the State Publishing House of Physical and Mathematical Literature after the death of its author. The book therefore appears without any alterations, if we except a few notes of a bibliographical character indicated by the letters (B. G.).

In spite of the fact that A. Ya. Khintchine wrote this book more than a quarter of a century ago, it has preserved all its charm and freshness. If is not without reason that this book has run to a large number of editions and a series of reprintings within the last ten years. More than this, in connection with the development of new methods in the technique of computation, critical interest has arisen in various computational algorithms, among these the algorithm of the continued fraction. In this direction there appeared, a few years ago, the useful monograph by A. N. Khovanskii ("The Application of Continued Fractions and their Generalisations to Problems in Approximation Theory", State Publishing House, 1956 [1]). Although A. Ya. Khintchine did not set himself a similar aim, his book can nevertheless serve as an excellent introduction both to the study of the algorithm of the continued fraction and to the deep and interesting problems of the metric theory of numbers, to the development of which the author contributed great strength and initiative. To a large extent the whole of the third chapter of this book is the result of his researches.

I hope that this small book will be read by as wide a circle of people and with the same enthusiasm that many, among them the author of these lines, read it twenty five years ago.

B. V. Gnedenko

[1]) English translation published by P. Noordhoff, Groningen, 1963 — *trans.*

Foreword to the Second Edition

The following second edition is a virtually unaltered version of the first.

Since the appearance of the first edition no further monographs concerned with continued fractions have appeared in the Russian literature. Among the general treatises on Number Theory which contain an elementary introduction to the theory of continued fractions, we mention the textbooks of D. A. Grave, B. A. Wenkoff and I. W. Arnold.

Moscow, October 1949 — A. KHINTCHINE

From the Foreword to the First Edition

The theory of continued fractions is concerned with a special algorithm which is one of the most important tools of Analysis, the Calculus of Probability, Mechanics and in particular of Number Theory. The aim of the following elementary textbook is to acquaint the reader with the so called *simplest* continued fraction, which has the form

$$a_0+\cfrac{1}{a_1+\cfrac{1}{a_2+\dots}}.$$

Here, moreover, it is assumed that all the elements a_i $(i \geqq 1)$ are positive integers. This most important and at the same time most extensively studied class of continued fractions lies at the root of almost all arithmetic and very many analytic applications of continued fractions.

I regard the publication of an elementary treatment of the theory of continued fractions as being necessary because this theory, which formerly belonged to the Courses of Mathematics in secondary

schools, is now no longer taught, and in consequence is no longer to be found in the textbooks of elementary algebra. On the other hand this theory is not part of the Institute of Technology programme (and indeed not even part of that of the University Mathematical Departments) and for this reason continued fractions are not to be found in the relevant Institute of Technology text books. Thus a specialist who wishes to learn something of the elements of the theory is compelled either to resort to old textbooks of the time before the Revolution or to foreign specialised treatises.

The main aim of this book is to close the gap in our literature described above; thus it is of necessity elementary and as compact as possible. By this reason its *style* is determined. Against this, its *content* slightly exceeds that which is absolutely necessary for the various applications. This is true above all with regard to the last chapter which is concerned with the fundamentals of the metric (or probability-theoretic) application of continued fractions. We are concerned here with an important new chapter which is virtually the exclusive creation of Soviet scientists. Many parts of Chapter II also go beyond the mentioned minimum, since I wished to discuss the fundamental rôle which continued fractions play in the investigation of irrational numbers, in as much detail as is possible within the framework of such an elementary introduction. If the theory of continued fractions is to be the subject of a special book, then it would seem to me futile to omit the very topics and interconnections of this theory which at the present time most occupy scientific thinkers.

As to the division of the material we need to point out that the "formal" part of the material is described in the introductory chapter. In this section the elements of the continued fraction are taken to be arbitrary positive (not necessarily whole) numbers and are assumed to be independent variables. A disadvantage of this scheme of development is that the formal properties of the apparatus are brought to the reader's attention before its concrete content. These properties are therefore dissociated from the content which, from a pedagogic point of view, would seem to be somewhat unfavorable.

But apart from the fact that in this way a greater methodological rigour is achieved (since the reader instantly recognises which properties of the continued fractions arise from the structure of the

apparatus, and which arise from the assumption that the elements are assumed to be positive integers) this provisional emphasis on the formalism allows the arithmetic theory, which is the principal concern of this book, to be built upon a completed foundation. Thus one can concentrate the reader's attention upon the concrete content of the material being explained, without having to distract him with considerations of formalism.

Moscow, 12 February 1935 A. KHINTCHINE

I. The Properties of the Calculus

§ 1 Introduction

The simplest continued fraction is given by an expression of the form

$$a_0+\cfrac{1}{a_1+\cfrac{1}{a_2+\ldots}}. \tag{1}$$

In the most general case the symbols $a_0, a_1, \ldots$ represent independent variables. Depending on the circumstances one can regard these variables as lying in any domain of definition. Thus one can assume that the $a_0, a_1, a_2, \ldots$ are real or complex numbers, functions of one or more variables, and so on. For the purposes of this book we shall assume that $a_1, a_2, \ldots$ are always *positive*, whilst a_0 is an *arbitrary real quantity*. We shall call these numbers the *elements* or *partial denominators* of the continued fraction under consideration. They can be either finite or infinite in number. In the first case we write the continued fraction in the form

$$a_0+\cfrac{1}{a_1+\cfrac{1}{a_2+\ldots}}\qquad\qquad\qquad

On the other hand we cannot immediately assign a numerical value to an infinite continued fraction. We can only speak of a formal expansion (if, that is, we impose no further conditions: the same situation exists when considering infinite series about whose convergence nothing has been said). Nevertheless it can still be the subject of a mathematical investigation.

For notational simplicity we shall agree from now on to write the infinite continued fraction (1) in the form

$$[a_0; a_1, a_2, \ldots], \tag{3}$$

and the terminating continued fraction (2) in the form

$$[a_0; a_1, a_2, \ldots, a_n]. \tag{4}$$

In this way the number of terms in a terminating continued fraction is equal to the number of symbols occurring in the square brackets after the semicolon.

We shall call the continued fraction

$$s_k = [a_0; a_1, a_2, \ldots, a_k]$$

with $0 \leqq k \leqq n$, a *section* of the continued fraction (4). In the same way we can refer to s_k, for any $k \geqq 0$, as being a section of the infinite continued fraction (3). Clearly any section of any (terminating or infinite) continued fraction is a terminating continued fraction.

Further we shall agree to call the continued fraction

$$r_k = [a_k; a_{k+1}, \ldots, a_n]$$

the *remainder* of the continued fraction (4). Similarly we shall refer to the continued fraction

$$r_k = [a_k; a_{k+1}, a_{k+2}, \ldots]$$

as the remainder of the infinite continued fraction (3). Clearly the remainders of all terminating continued fractions in turn are terminating continued fractions, whilst all remainders of an infinite continued fraction are infinite continued fractions.

By definition it follows that for a terminating continued fraction,

$$[a_0; a_1, a_2, \ldots, a_n] = [a_0; a_1, a_2, \ldots, a_{k-1}, r_k] \quad (0 \leqq k \leqq n). \tag{5}$$

The analogous relationship

$$[a_0; a_1, a_2, \ldots] = [a_0; a_1, a_2, \ldots, a_{k-1}, r_k] (k \geqq 0)$$

for infinite continued fractions has so far only a formal meaning, since the element r_k on the right hand side of this equation has no defined numerical value.

§ 2 Convergents

Since every terminating continued fraction

$$[a_0; a_1, a_2, \ldots, a_n]$$

is the result of a finite number of rational operations upon its elements, it is a rational function of these elements and may be written as the quotient

$$\frac{P(a_0, a_1, \ldots, a_n)}{Q(a_0, a_1, \ldots, a_n)}$$

of two polynomials with integer coefficients in the $a_0, a_1, \ldots a_n$. If the elements are given numerical values, then the continued fraction itself may be written as a simple fraction p/q. Clearly, however, such an expression would not be unique. In the following it will be important to be able to give a *definite* expression for a terminating continued fraction in the form of a simple fraction. We shall call this representation *canonical*, and define it inductively.

We choose, as the canonical representation of the 0-termed continued fraction

$$[a_0] = a_0$$

the fraction $a_0/1$. Now suppose that the canonical representations of continued fractions whose elements are less than n in number have been defined. Using formula (5) we may express the n-termed continued fraction $[a_0; a_1, a_2, \ldots, a_n]$ as

$$[a_0; a_1, \ldots, a_n] = [a_0; r_1] = a_0 + \frac{1}{r_1}.$$

Here

$$r_1 = [a_1; a_2, \ldots, a_n]$$

is an $(n-1)$-termed continued fraction for which a canonical representation already exists. Let it have the form

$$r_1 = \frac{p'}{q'};$$

then

$$[a_0; a_1, a_2, \ldots, a_n] = a_0 + \frac{q'}{p'} = \frac{a_0 p' + q'}{p'}.$$

We shall use this last fraction as the canonical representation of the continued fraction $[a_0; a_1, \ldots, a_n]$. Putting

$$[a_0; a_1, \ldots, a_n] = \frac{p}{q}$$

$$r_1 = [a_1; a_2, \ldots, a_n] = \frac{p'}{q'}$$

we obtain for the numerator and denominator of this canonical representation, the relationships

$$p = a_0 p' + q', \quad q = p'. \tag{6}$$

At the same time we see that the canonical representations of terminating continued fractions having arbitrarily many terms are now uniquely determined.

The canonical representations of the sections of a given (terminating or infinite) continued fraction $\alpha = [a_0; a_1, a_2, \ldots]$ play a particularly important role in the theory of continued fractions.

We shall denote the canonical representation of the section

$$s_k = [a_0; a_1, a_2, \ldots, a_k]$$

of α by p_k/q_k, and call it the *convergent of order k* of α. This definition holds for both terminating and infinite continued fractions. The only difference is that a terminating continued fraction possesses a finite number of convergents, whilst in the case of an infinite continued fraction the number of convergents is infinite. Evidently, for a terminating continued fraction,

$$\frac{p_n}{q_n} = \alpha.$$

Such a continued fraction has in all $n+1$ convergents (of orders $0, 1, \ldots, n$).

THEOREM 1. (Law of formation of the convergents). *For any* $k \geqq 2$

$$\left.\begin{aligned} p_k &= a_k p_{k-1} + p_{k-2}, \\ q_k &= a_k q_{k-1} + q_{k-2}. \end{aligned}\right\} \tag{7}$$

PROOF. In the case $k = 2$ the formulae (7) may easily be verified directly. Let us suppose that they are valid for all $k < m$. Now we consider the continued fraction

$$[a_1; a_2, \ldots, a_n]$$

and denote its rth order convergent by p'_r/q'_r. From (6)

$$\begin{aligned} p_n &= a_0 p'_{n-1} + q'_{n-1} \\ q_n &= p'_{n-1} \end{aligned}$$

and by assumption

$$\begin{aligned} p'_{n-1} &= a_n p'_{n-2} + p'_{n-3}, \\ q'_{n-1} &= a_n q'_{n-2} + q'_{n-3} \end{aligned}$$

(note that a_n occurs here and not a_{n-1}, since the continued fraction $[a_1; a_2, \ldots, a_n]$ begins with a_1, not with a_0). Hence we obtain from formula (6)

$$\begin{aligned} p_n &= a_0(a_n p'_{n-2} + p'_{n-3}) + (a_n q'_{n-2} + q'_{n-3}) \\ &= a_n(a_0 p'_{n-2} + q'_{n-2}) + (a_0 p'_{n-3} + q'_{n-3}) \\ &= a_n p_{n-1} + p_{n-2}, \\ q_n &= a_n p'_{n-2} + p'_{n-3} = a_n q_{n-1} + q_{n-2}, \end{aligned}$$

which was to be proved.

The recursion formulae (7), which express the numerator and denominator of the nth order convergent in terms of a_n and the numerators and denominators of the preceding two convergents, serve as a formal basis for the whole theory of continued fractions.

REMARK. It will be convenient in the following to bring the convergent of order -1 into the scheme of things, and for this we put $p_{-1} = 1$ and $q_{-1} = 0$. Under this assumption (and only then) formulae (7) are also valid for $k = 1$.

THEOREM 2. *For all* $k \geqq 0$

$$q_k p_{k-1} - p_k q_{k-1} = (-1)^k. \tag{8}$$

PROOF. Multiply equations (7) by q_{k-1} and q_k respectively and subtract the second from the first; we obtain

$$q_k p_{k-1} - p_k q_{k-1} = -(q_{k-1} p_{k-2} - p_{k-1} q_{k-2}).$$

Since

$$q_0p_{-1}-p_0q_{-1}=1$$

the proof of the theorem follows.

COROLLARY. *For all* $k \geqq 1$

$$\frac{p_{k-1}}{q_{k-1}}-\frac{p_k}{q_k}=\frac{(-1)^k}{q_kq_{k-1}}. \tag{9}$$

THEOREM 3. *For all* $k \geqq 1$

$$q_kp_{k-2}-p_kq_{k-2}=(-1)^{k-1}a_k.$$

PROOF. Multiply equations (7) by q_{k-2} and p_{k-2} respectively and subtract the first from the second. Using Theorem 2, we obtain

$$q_kp_{k-2}-p_kq_{k-2}=a_k(q_{k-1}p_{k-2}-p_{k-1}q_{k-2})=(-1)^{k-1}a_k,$$

which was to be proved.

COROLLARY. *For all* $k \geqq 2$

$$\frac{p_{k-2}}{q_{k-2}}-\frac{p_k}{q_k}=\frac{(-1)^{k-1}a_k}{q_kq_{k-2}}. \tag{10}$$

The simple results which we have just obtained allow us to draw some extremely important conclusions concerning the distribution of the convergents. Indeed equation (10) tells us that the even order convergents form an increasing sequence, the odd order convergents a decreasing sequence, so that the two sequences approach each other (all this is of course only valid under the assumption which we made that all elements from a_1 onwards are positive). Since, by virtue of equation (9), every odd order convergent is greater than the even order convergent which immediately follows it, it clearly follows that any odd order convergent is greater than every even order convergent. Hence we can draw the following conclusion.

THEOREM 4. *The convergents of even order form an increasing sequence, the convergents of odd order a decreasing sequence. Furthermore every odd order convergent is greater in value than every even order convergent.*

In particular, for a terminating continued fraction, every even order convergent is less than α and every odd order convergent

greater than α (the last convergent is exceptional, being equal to α).

To conclude this section we prove two simple but nevertheless extremely important theorems concerning the numerators and denominators of the convergents.

THEOREM 5. *For any* k $(1 \leqq k \leqq m)$

$$[a_0; a_1, a_2, \ldots, a_n] = \frac{p_{k-1}r_k + p_{k-2}}{q_{k-1}r_k + q_{k-2}} \tag{11}$$

(here p_i, q_i and r_i refer to the continued fraction on the left hand side of this equation).

PROOF. From (5) it follows that

$$[a_0; a_1, a_2, \ldots, a_n] = [a_0; a_1, a_2, \ldots, a_{k-1}, r_k].$$

Clearly the convergent of order $k-1$ of the continued fraction on the right hand side of this equation is equal to the fraction p_{k-1}/q_{k-1}. Its kth order convergent, p_k/q_k, is the value of the continued fraction. However, from formulae (7)

$$p_k = p_{k-1}r_k + p_{k-2}, \qquad q_k = q_{k-1}r_k + q_{k-2}$$

thus the theorem has been proved.

THEOREM 6. *For any* $k \geqq 1$

$$\frac{q_k}{q_{k-1}} = [a_k; a_{k-1}, \ldots, a_1].$$

PROOF. When $k = 1$ this relationship is clearly correct, for it then has the form

$$\frac{q_1}{q_0} = a_1.$$

Now let $k > 1$ and suppose that it has already been proved that

$$\frac{q_{k-1}}{q_{k-2}} = [a_{k-1}; a_{k-2}, \ldots, a_1]. \tag{12}$$

From the second of equations (7)

$$q_k = a_k q_{k-1} + q_{k-2},$$

and we obtain

$$\frac{q_k}{q_{k-1}} = a_k + \frac{q_{k-2}}{q_{k-1}} = \left[a_k; \frac{q_{k-1}}{q_{k-2}}\right],$$

and from this, using (5) and (12)

$$\frac{q_k}{q_{k-1}} = [a_k; a_{k-1}, \ldots, a_1]$$

which was to be proved.

§ 3 Infinite Continued Fractions

To every infinite continued fraction

$$[a_0; a_1, a_2, \ldots] \tag{13}$$

there corresponds an infinite sequence of convergents

$$\frac{p_0}{q_0}, \frac{p_1}{q_1}, \ldots, \frac{p_k}{q_k}, \ldots. \tag{14}$$

Every convergent represents a real number. In the event that the sequence (14) converges, i.e. possesses a limiting value α, it is eminently reasonable to regard α as being the value of the continued fraction; one writes in this case

$$\alpha = [a_0; a_1, a_2, \ldots].$$

One says, in this case, that the continued fraction (13) is *convergent*. If, however, the sequence (14) has no limit, then one says that the continued fraction (13) is *divergent*.

Convergent continued fractions display in their properties many analogies to terminating continued fractions. Their main property, which allows us to develop this analogy very far, is based on the following theorem.

THEOREM 7. *If the continued fraction* (13) *converges, then all its remainders converge. Conversely, if at least one remainder of the continued fraction* (13) *converges, then the continued fraction itself converges.*

PROOF. Let us denote by p_k/q_k the convergents of the given infinite continued fraction, and by p_k'/p_k' those of any one of its remainders r_n.

From formulae (11) we clearly have

$$\frac{p_{n+k}}{q_{n+k}} = [a_0; a_1, a_2, \ldots, a_{n+k}] = \frac{p_{n-1}\dfrac{p'_k}{q'_k} + p_{n-2}}{q_{n-1}\dfrac{p'_k}{q'_k} + q_{n-2}} \quad (k = 0, 1, \ldots). \tag{15}$$

If the remainder r_n converges, i.e. if the fraction p'_k/q'_k possesses a limit (which we also call r_n) as $k \to \infty$, then it follows immediately that the fraction p_{n+k}/q_{n+k} also tends to a limit α, whose value is given by

$$\alpha = \frac{p_{n-1}r_n + p_{n-2}}{q_{n-1}r_n + q_{n-2}}. \tag{16}$$

Solving equation (15) for p'_k/q'_k, then we can persuade ourselves in the same way that the converse assertion is true; thus Theorem 7 has been proved.

We point out that formula (16) which we have just proved for convergent infinite continued fractions, is completely analogous to formula (11) which we found earlier for terminating continued fractions. Thus for a convergent infinite continued fraction a theorem is valid which corresponds to Theorem 5 (which relates to terminating continued fractions).

From Theorem 4 of the previous section we clearly have for convergent infinite continued fractions the following

THEOREM 8. *The value of a convergent infinite continued fraction is greater than that of any even order convergent and less than that of any odd order convergent.*

Furthermore, as a consequence of Theorem 2 of the previous section we derive a result which, because of Theorem 8, plays an important rôle in the arithmetic application of the theory of continued fractions.

THEOREM 9. *The value* α *of the infinite continued fraction* (13) *satisfies the inequality* [1])

$$\left|\alpha - \frac{p_k}{q_k}\right| < \frac{1}{q_k q_{k+1}}$$

for any $k \geqq 0$.

[1]) It should be pointed out that under the assumptions which we have made, $q_k > 0$ for all $k \geqq 0$. For $q_0 = 1$ and $q_1 = a_1$: from these it follows by an inductive proof based on the second of equations (7) that $q_k > 0$ for all $k > 1$.

Clearly Theorem 9 is also valid for the terminating continued fraction

$$\alpha = [a_0; a_1, a_2, \ldots, a_n]$$

for all $k < n$, where only for $k = n-1$ does the inequality become an identity (since in this case $p_n/q_n = \alpha$).

If α is the value of a convergent infinite continued fraction, then we shall refer to the elements of this continued fraction as the elements of the number α. Similarly we shall regard the convergents, sections, and remainders of the continued fraction (13) as being the convergents, sections, and remainders of the number α. From theorem 7 all remainders of a convergent infinite continued fraction have well-determined real values.

When considering infinite continued fractions, questions concerning convergence arise quite naturally, just as they do in the case of infinite series. In the case being considered here, i.e. where $a_i > 0$ for all $i > 1$, a very simple and convenient convergence criterium can be given.

THEOREM 10. *For the continued fraction* (13) *to converge, it is necessary and sufficient that the series*

$$\sum_{n=1}^{\infty} a_n \tag{17}$$

should diverge.

PROOF. From Theorem 4 it is evident that for an infinite continued fraction to converge it is necessary and sufficient that the two sequences considered in this theorem should have the same limit (that each of the two sequences should possess a limit follows from the theorem in any case). This follows, however, as is shown by formula (9), if and only if

$$q_k q_{k+1} \to \infty \text{ for } k \to \infty. \tag{18}$$

The condition (18) is thus necessary and sufficient for the convergence of the continued fraction in question.

Now let the series (17) converge. From the second of formulae (7),

$$q_k > q_{k-2}. \qquad (k \geqq 1).$$

Thus for any k either $q_k > q_{k-1}$ or $q_{k-1} > q_{k-2}$. In the first case the second of formulae (7) yields

$$q_k < a_k q_k + q_{k-2}$$

from which, for sufficiently large k,

$$q_k < \frac{q_{k-2}}{1-a_k}$$

(that is, if $a_k < 1$; from the convergence of the series (17) this must be so for some $k \geqq k_0$). In the second case this formula yields (for $a_k < 1$)

$$q_k < (1+a_k) q_{k-1} < \frac{q_{k-1}}{1-a_k}.$$

Thus for all $k \geqq k_0$ we obtain

$$q_k < \frac{1}{1-a_k} q_l$$

with $l \leqq k$. If now $l \geqq k_0$, the same inequality may be applied to q_l. Continuing this reasoning, we clearly arrive at the following inequality

$$q_k < \frac{q_s}{(1-a_k)(1-a_l)\ldots(1-a_r)} \tag{19}$$

Here $k > l > \ldots > r > k_0$ and $s \leqq k_0$. From the assumption that the series (17) converges, it is known that the infinite product

$$\prod_{n=k_0}^{\infty} (1-a_n)$$

also converges, i.e. it possesses a positive limit which we shall denote by λ. Clearly

$$(1-a_k)(1-a_l)\ldots(1-a_r) \geqq \prod_{n=k_0}^{k} (1-a_n) = \lambda.$$

Therefore, denoting by Q the largest of the numbers $q_0, q_1, \ldots, q_{k_0-1}$, we can conclude from the inequality (19) that

$$q_k < \frac{Q}{\lambda} \qquad (k \geqq k_0).$$

Consequently

$$q_{k+1} q_k < \frac{Q^2}{\lambda^2} \qquad (k \geqq k_0)$$

and relationship (18) cannot be true. In this case, however, the given continued fraction must be divergent.

Now let the series (17) be divergent. Since $q_k > q_{k-2}$ for all $k \geqq 2$, we obtain (denoting the smaller of the numbers q_0 and q_1 by c) the inequality $q_k \geqq c$ for any $k \geqq 0$. Thus the second of formulae (7) yields

$$q_k \geqq q_{k-2} + ca_k \qquad (k \geqq 2).$$

By repeated application of this inequality we obtain

$$q_{2k} \geqq q_0 + c \sum_{n=1}^{k} a_{2n}$$

and

$$q_{2k+1} \geqq q_1 + c \sum_{n=1}^{k} a_{2n+1}$$

from which

$$q_{2k} + q_{2k+1} > q_0 + q_1 + c \sum_{n=1}^{2k+1} a_n,$$

or in other words

$$q_k + q_{k-1} > c \sum_{n=1}^{n} a_n$$

for all k. We have proved this inequality for odd k, but it is manifest that it can be derived for even k in the same way.

From this it follows that at least one of the factors in the product $q_k q_{k-1}$ is greater than $\frac{c}{2} \sum_{n=1}^{k} a_n$. Since, however, the other factor cannot be less than c, we obtain

$$q_k q_{k-1} > \frac{c^2}{2} \sum_{n=1}^{k} a_n.$$

Relationship (18), and hence the convergence of the given continued fraction, follows from the assumed divergence of the series (17). Thus Theorem 10 has been proved in its entirety.

§ 4 Continued Fractions with Natural Elements

From this section to the end of the book we shall assume that the elements of any given continued fractions are *natural*, i.e. are positive integers, a_0 will also be an integer, not however, necessarily

positive. If such a continued fraction is infinite, then from Theorem 10 it is always convergent. We can thus safely assume in the following, without any further indication, that any given infinite continued fraction is convergent; we can thus speak of its *value* or *magnitude*.

If such a continued fraction is terminating, and if its last element $a_n = 1$, then evidently $r_{n-1} = a_{n-1}+1$—an integer. In this case we can therefore write the given continued fraction $[a_0; a_1, a_2, \ldots, a_{n-1}, 1]$ in the form of an $(n-1)$-termed continued fraction $[a_0; a_1, a_2, \ldots a_{n-1}+1]$ in which, in this form, the last element is greater than 1.

From the fact that we have just established, it follows that we can except from our considerations all those terminating continued fractions whose last element is equal to 1 (apart, that is, from the zero-termed continued fraction [1]). This remark is extremely important for the uniqueness of the representation of numbers by means of continued fractions (see Chapter II, § 5).

Clearly, in the case under consideration, the numerators and denominators of the convergents are integers (for p_{-1}, q_{-1}, p_0, q_0 this is immediately visible; for the rest it follows from formulae (7)). Apart from this we have the following most important

THEOREM 11. *All convergents are irreducible.*

PROOF. The theorem follows immediately from Theorem 8, since any common factor of p_n and q_n is also a factor of the expression $q_n p_{n-1} - p_n q_{n-1}$.

The second of formulae (7) indicates that $q_k > q_{k-1}$ for all $k \geqq 2$. Thus the sequence

$$q_1, q_2, \ldots, q_k, \ldots$$

increases monotonically. As to the nature of this increase, a far sharper statement can be made.

THEOREM 12. *For any* [1] $k \geqq 2$

$$q_k \geqq 2^{(k-1)/2}.$$

---

[1]) Here and in the following it will be understood that in the case of a *terminating* continued fraction only such values of k are considered for which q_k has meaning.

PROOF. For $k \geqq 2$

$$q_k = a_k q_{k-1} + q_{k-2} \geqq q_{k-1} + q_{k-2} \geqq 2q_{k-2}.$$

By successive application of this inequality we obtain

$$q_{2k} \geqq 2^k q_0 = 2^k, \; q_{2k+1} \geqq 2^k q_1 \geqq 2^k;$$

thus our theorem is clearly proved.

The denominators of the convergents therefore do not increase more slowly than the terms of a certain geometric series.

Intermediate fractions. Let $k \geqq 2$ and i be an arbitrary non-negative whole number. The difference

$$\frac{p_{k-1}(i+1)+p_{k-2}}{q_{k-1}(i+1)+q_{k-2}} - \frac{p_{k-1}i+p_{k-2}}{q_{k-1}i+q_{k-2}}$$

is, as one can easily verify, equal to

$$\frac{(-1)^k}{[q_{k-1}(i+1)+q_{k-2}][q_{k-1}i+q_{k-2}]};$$

for all $i \geqq 0$ it has the same sign, which only depends upon whether k is even or odd. From this it follows that the fractions

$$\frac{p_{k-2}}{q_{k-2}}, \frac{p_{k-2}+p_{k-1}}{q_{k-2}+q_{k-1}}, \frac{p_{k-2}+2p_{k-1}}{q_{k-2}+2q_{k-1}}, \ldots, \frac{p_{k-2}+a_k p_{k-1}}{q_{k-2}+a_k q_{k-1}} = \frac{p_k}{q_k} \quad (20)$$

constitute an increasing sequence for even k, and a decreasing sequence for odd k (see Theorem 4). The bounding members of this sequence are convergents which are either both of even order or both of odd order. We shall call the terms of the sequence standing between them (insofar as they exist, i.e. if $a_k > 1$) *intermediate fractions*. They play a significant rôle in arithmetic applications; they are not, however, quite as important as the convergents. In order better to understand their distribution and the law of their successive formation, it will be convenient to introduce the concept of the so called *median value* of two fractions.

The median value of two fractions a/b and c/d with positive denumerators is given by the fraction

$$\frac{a+c}{b+d}.$$

LEMMA. *The value of the median of two fractions always lies between those of the two fractions.*

PROOF. For definiteness let $a/b \leqq c/d$; then $bc-ad \geqq 0$ and consequently

$$\frac{a+c}{b+d} - \frac{a}{b} = \frac{bc-ad}{b(b+d)} \geqq 0, \quad \frac{a+c}{b+d} - \frac{c}{d} = \frac{ad-bc}{b(b+d)} \leqq 0,$$

which was to be proved.

We recognise immediately that every intermediate fraction of the sequence (20) is the median value of the previous fraction and the fraction p_{k-1}/q_{k-1}. We start with the convergent p_{k-2}/q_{k-2} and proceed further in the sequence (20) by means of forming median values. At the last step the median value derived is the convergent p_k/q_k. Thus this lies between p_{k-1}/q_{k-1} and p_{k-2}/q_{k-2}, which, of course, we already knew from Theorem 4. Moreover, we know that the value α of the given continued fraction lies between p_{k-1}/q_{k-1} and p_k/q_k, and that the convergents p_{k-2}/q_{k-2} and p_k/q_k whose orders are either both even or both odd, are either both larger or both smaller than α. From this it follows that the members of the sequence (20) are either all less than or all greater than α. In the first case the fraction p_{k-1}/q_{k-1} is greater, in the second case less, than α. In particular the fractions $(p_{k-1}+p_{k-2})/(q_{k-1}+q_{k-2})$ and p_{k-1}/q_{k-1} always lie on opposite sides of α. *The value of a continued fraction always lies between those of any convergent and the median value formed from this and the previous convergent.* (We recommend to the reader that he should make a schematic diagram which shows how all these numbers are distributed).

This remark provides us with a method for constructing the next convergent p_k/q_k, as soon as we know the convergent p_{k-2}/q_{k-2} and p_{k-1}/q_{k-1}. The element a_k need not be known; however, we must know the value α of the continued fraction itself. In the event, we need only to form the median value of the two given fractions, then the median value of this median value and p_{k-1}/q_{k-1}, and so on, always forming the median value of the median value we have just previously derived and p_{k-1}/q_{k-1}. We know that the successive median values approach α. *The last median value of this sequence lying on the same side of* α *as* p_{k-2}/q_{k-2} *(from which we started) is the required convergent* p_k/q_k. Indeed we know already that p_k/q_k is available among the median values which were formed, and that

it lies on the same side of α as p_{k-2}/q_{k-2}. We have therefore only to show that the very next median value lies on the other side of α; the next median value, however, is $(p_k+p_{k-1})/(q_k+q_{k-1})$ and from the remark which was made earlier, is known to lie on the other side of α.

A further, and even more important consequence of the relative positioning of α and its convergents and intermediate fractions, arises from the following considerations. The intermediate fraction $(p_k+p_{k-1})/(q_k+q_{k-1})$ is enclosed between p_k/q_k and α and lies nearer to p_k/q_k than does α, i.e.

$$\left|\alpha-\frac{p_k}{q_k}\right|>\left|\frac{p_k+p_{k+1}}{q_k+q_{k+1}}-\frac{p_k}{q_k}\right|=\frac{1}{q_k(q_k+q_{k+1})}$$

(an equality sign is not possible here; for in this case

$$\alpha=\frac{p_k+p_{k+1}}{q_k+q_{k+1}}=\frac{p_{k+2}}{q_{k+2}},\quad a_{k+2}=1,$$

i.e. α would be a terminating continued fraction with last element 1; but at the beginning we specifically excluded consideration of such continued fractions).

In this way we obtain the following important

THEOREM 13. *For all* $k\geqq 0$

$$\left|\alpha-\frac{p_k}{q_k}\right|>\frac{1}{q_k(q_{k+1}+q_k)}. \tag{21}$$

The inequality (21), which yields a *lower bound* for the difference $\left|\alpha-\frac{p_k}{q_k}\right|$, is evidently a complement to the inequality obtained in Theorem 9, which gives an upper bound for the same difference.

II. The Representation of Numbers by means of Continued Fractions

§ 5 The Continued Fraction as an Instrument for Representing Real Numbers

Theorem 14. *To every real number α there corresponds, uniquely, a continued fraction whose value is this number. This continued fraction terminates if α is rational. If α is irrational, it is infinite* [1]).

Proof. Denote by a_0 the largest integer which does not exceed α. If α is not an integer, then we may write

$$\alpha = a_0 + \frac{1}{r_1} \tag{22}$$

to determine the number r_1. Clearly $r_1 > 1$, for

$$\frac{1}{r_1} = \alpha - a_0 < 1.$$

In general we shall denote by a_n the largest integer which does not exceed r_n (if, that is, r_n is not an integer); the number r_{n+1} is determined from the relationship

$$r_n = a_n + \frac{1}{r_{n+1}}. \tag{23}$$

This process can clearly be continued until some number r_n is a whole number; from this it is clear that $r_n > 1$ $(n \geqq 1)$.

Equation (22) shows that

$$\alpha = [a_0; r_1].$$

Now, in general, let

$$\alpha = [a_0; a_1, a_2, \ldots, a_{n-1}, r_n]. \tag{24}$$

Then there follows from relationship (23) and formula (5) of

[1]) We recall that we are considering continued fractions whose elements are integers with $a_i > 0$ for $i \geqq 1$, and that the last element of any terminating continued fraction differs from unity.

Chapter I

$$\alpha = [a_0; a_1, a_2, \ldots, a_{n-1}, a_n, r_{n+1}].$$

Thus formula (24) is valid for all n (clearly it is assumed that the $r_1, r_2, \ldots, r_{n-1}$ are not integers).

If the number α is rational, then clearly so are all the r_n. It is easy to see that in this case our process will come to a stop after a finite number of steps. If, for example, $r_n = a/b$, then in the event

$$r_n - a_n = \frac{a - ba_n}{b} = \frac{c}{b}$$

with $c < b$, since $r_n - a_n < 1$. Equation (23) yields

$$r_{n+1} = \frac{b}{c}$$

(to the extent that c is not equal to zero, i.e. that r_n is not an integer; but in this case our assertion is already proved). From this r_{n+1} has a smaller denominator than r_n; thus after a finite number of steps in the sequence $r_1, r_2, \ldots$ we must come to an integer $r_n = a_n$. In this case, however, formula (24) tells us that the number α is represented by a terminating continued fraction whose last element is $a_n = r_n > 1$.

If the number α is irrational, then the r_n are also, and our process can be continued without limit. Writing

$$[a_0; a_1, a_2, \ldots, a_n] = \frac{p_n}{q_n}$$

where the numbers p_n and q_n are coprime, and $q_n > 0$, we obtain from formula (24) and formula (16) of Chapter I

$$\alpha = \frac{p_{n-1} r_n + p_{n-2}}{q_{n-1} r_n + q_{n-2}} \qquad (n \geqq 2).$$

On the other hand, manifestly,

$$\frac{p_n}{q_n} = \frac{p_{n-1} a_n + p_{n-2}}{q_{n-1} a_n + q_{n-2}},$$

whence

$$\alpha - \frac{p_n}{q_n} = \frac{(p_{n-1} q_{n-2} - q_{n-1} p_{n-2})(r_n - a_n)}{(q_{n-1} r_n + q_{n-2})(q_{n-1} a_n + q_{n-2})},$$

and further

$$\left|\alpha-\frac{p_n}{q_n}\right|<\frac{1}{(q_{n-1}r_n+q_{n-2})(q_{n-1}a_n+q_{n-2})}<\frac{1}{q_n^2}.$$

Thus, as $n\to\infty$

$$\frac{p_n}{q_n}\to\alpha;$$

this clearly means that the value of the infinite continued fraction $[a_0; a_1, a_2, \ldots]$ is α.

In this way we have shown that the number α can always be represented by a continued fraction. This terminates if α is rational; it is infinite if α is irrational. It only remains for us to show that the development obtained is unique. We point out that this uniqueness already follows from the remarks in § 4 of Chapter I; there we saw that we may successively derive all convergents, and consequently all elements, as soon as we know the value of the continued fraction. But the required uniqueness can be established in a far simpler way. Let

$$\alpha=[a_0; a_1, a_2, \ldots]=[a_0'; a_1', a_2', \ldots],$$

where these continued fractions can be either infinite or terminating. We denote by $[x]$ the largest integer not exceeding x. Now it is clear that $a_0=[\alpha]$ and $a_0'=[\alpha]$, from which $a_0=a_0'$. Suppose that we have already established that

$$a_i=a_i', \qquad (i=0, 1, \ldots, n)$$

then, using a self evident notation,

$$\left.\begin{aligned}p_i&=p_i'\\ q_i&=q_i'\end{aligned}\right\} \qquad (i=0, 1, \ldots, n)$$

and from (16)

$$\alpha=\frac{p_nr_{n+1}+p_{n-1}}{q_nr_{n+1}+q_{n-1}}=\frac{p_n'r_{n+1}'+p_{n-1}'}{q_n'r_{n+1}'+q_{n-1}'}=\frac{p_nr_{n+1}'+p_{n-1}}{q_nr_{n+1}'+q_{n-1}}.$$

From this it follows that $r_{n+1}=r_{n+1}'$, and since both $a_{n+1}=[r_{n+1}]$ and $a_{n+1}'=[r_{n+1}']$, also $a_{n+1}=a_{n+1}'$, i.e. the two given continued fractions are identically equal, which was to be proved.

This reasoning does not, however, hold true if we had allowed

ourselves to consider terminating continued fractions whose last element is equal to unity. If for example $a_{n+1} = 1$ were to be such a last element then $r_n = a_n + 1$ and $a_n \neq [r_n]$.

We have now shown that real numbers can be uniquely represented by means of continued fractions. The principal value of such a representation clearly lies in the consideration that as soon as we know that a continued fraction represents a certain number, we can compute this number to any prescribed accuracy. For this reason the calculus of continued fractions may, at least in principle, make the same claims for consideration as those which may be made on behalf of decimal fractions, or quite generally for systematic fractions (i.e. those which use any fixed radix system).

Now what are the essential advantages and disadvantages of continued fractions as a means of representing real numbers when compared with the far more widely established systematic fractions? In order to be able to answer this question, one must, before anything else, be absolutely clear as to what can and what must be demanded of a calculus of this sort. Clearly the first and most essential *theoretical* requirement consists in that the apparatus should be able to reflect, as completely as is possible, the properties of the number that it represents, and to do so in such a way that these properties can be recognised as easily as possible as soon as the representation of the numbers by means of this apparatus is given.

With regard to this first requirement, the preference must without doubt be conceded to continued fractions as opposed to systematic fractions (this is especially true with regard to decimal fractions). We shall be able to convince ourselves of this as we progress through this book. To a certain extent this is already clear from the following a priori considerations: whilst every systematic fraction is coupled to a definite radix system and therefore unavoidably reflects more the interaction of the radix system and the number, than the absolute properties of the number itself, the continued fraction is completely independent of any radix system and reproduces in a pure form the properties of the number which it represents. We have already seen that the fact that a number is rational or irrational is indicated by the fact that its continued fraction representation either terminates or is infinite. As is known the corresponding phenomenon is more difficult to recognise in the case of systematic fractions:

whether the fraction which represents a number terminates or not depends, apart from the properties of the number itself, much more upon the relationship between the number and the radix system.

Apart from the fundamental requirements which we have mentioned, we must demand of any calculus which serves for the representation of numbers that it should also satisfy certain *practical* requirements (certain of which may also be of theoretical significance). For example, it is essential, using the calculus in question, to be able to determine to any prescribed accuracy the approximate value of the number which is being represented, and to be able to do this as simply as possible. This requirement is satisfied by continued fractions to a very high degree, and in any case far better than by systematic fractions. Furthermore, we shall very soon see that the convergents which are yielded by the continued fraction display a property which in a certain extremely simple and important sense, is that of *best approximation*.

There is however, another, and far more important, practical requirement which continued fractions in no way satisfy. The necessities of computation mean that it is desirable that associated with any representational apparatus the knowledge of a few number representations should permit the representation of the simplest functions (most importantly, the sum and product) of these numbers to be found sufficiently simply. In other words an apparatus which is suitable from a practical point of view must offer the possibility of interpreting *arithmetic operations* in terms of the simplest possible rules; otherwise it cannot serve as an aid to computation. It is well known how convenient systematic fractions are in this connection. As opposed to this these are no simple rules for arithmetic operations upon continued fractions. Even the task of finding the continued fraction development of the sum of two continued fractions is extremely complicated and in practice incapable of being carried out.

The advantages and shortcomings of continued fractions as opposed to systematic fractions which have just been described, determine to a considerable extent the bounds of the domains in which the two representational apparatuses find their application. Whilst in practical calculations systematic fractions are used almost exclusively, use of continued fractions is to be preferred for the

theoretical investigation of the arithmetic laws of the continuum and the arithmetic properties of certain kinds of irrational numbers, and is the best and indispensable tool for this sort of inquiry. The main task of the following work consists precisely in establishing the significance for this purpose of the mechanism of the continued fraction.

§ 6 Convergents as Best Approximations

If we wish to represent (to a certain accuracy) an irrational number α by means of a vulgar fraction, we can quite naturally make use of the convergents of the continued fraction which represents this number. The degree of approximation which can be attained in this way is determined by Theorems 9 and 13 of Chapter I: indeed

$$\frac{1}{q_n(q_n+q_{n+1})} < \left|\alpha - \frac{p_n}{q_n}\right| \leqq \frac{1}{q_n q_{n+1}}.$$

The problem of approximating an irrational number by means of a rational fraction is, in its simplest form, usually posed in the following way: we look for that rational fraction having the smallest (positive) denominator which does not differ from the given irrational number by a given amount. The problem posed is also meaningful if α is a rational number. For example if α is a fraction whose numerator and denominator are large numbers, it might be necessary to represent this number approximately by means of a fraction whose numerator and denominator are smaller numbers. From the purely practical point of view the two cases (rational and irrational α) are indistinguishable; since in practice every number is only given to a certain accuracy.

One can immediately see that systematic fractions are quite inappropriate for the solution of this problem, since the denominators of the approximating fractions which arise depend exclusively upon the chosen radix system (in the case of decimal fractions they are powers of 10) and not at all upon the arithmetic nature of the number to be represented. As opposed to this, in the case of continued fractions, the denominators of the convergents are completely determined by the number to be represented. We therefore have every reason to assume that these convergents which are in the most

natural way coupled with number to be represented and are uniquely determined by it, should play an essential rôle in the problem of finding the best approximation to this number by means of a rational fraction [1]).

We shall say that the rational fraction $a/b (b > 0)$ is a *best approximation* to the real number α if every other rational fraction having the same or smaller denominator differs from this number more than a/b, in other words; if for $0 < d \leqq b$, $a/b \neq c/d$ then

$$\left|\alpha - \frac{c}{d}\right| > \left|\alpha - \frac{a}{b}\right|.$$

THEOREM 15. *Every best approximation to the numbers α is either a convergent or an intermediate fraction of the continued fraction which represents this number.*

PRELIMINARY REMARK: If this theorem is to be valid without exception then we must (as we agreed to do in § 2) take into account convergents of order -1. Indeed, for example, the fraction $\frac{1}{3}$ is a best approximation to the number $\frac{1}{4}$ but is neither a convergent nor an intermediate fraction, since the set of these fractions (that is if the convergents are to begin with that of order 0) contains only the two numbers $\frac{0}{1}$ and $\frac{1}{4}$. If, however, one inserts the fraction $\frac{1}{0}$ as convergent of order -1, then one obtains the sequence

$$\frac{0}{1}, \frac{1}{0}, \frac{1}{1}, \frac{1}{2}, \frac{1}{3}, \frac{1}{4},$$

as the set of convergents and intermediate fractions, and this contains the fraction $\frac{1}{3}$.

PROOF. Let a/b be a best approximation to the number α. Then $a/b \geqq a_0$. Indeed in the case that $a/b < a_0$, the fraction $a_0/1$ (which differs from a/b and whose numerator is not greater than b) would lie closer to α than a/b. But then a/b would not be a best approximation.

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[1]) Two interesting algorithms for representing irrational numbers were suggested by M. V. Ostrogradski shortly before his death. His brief remarks on the calculations were contained in papers left with the manuscript depository of the Academy of Sciences of the Ukrainian S.S.P. These notes were incorporated in the article by E. Ya. Remez "On series with alternating signs, which may be related to two algorithms of M. V. Ostrogradski for the approximation of irrational numbers" Advances in Mathematical Science, 6, 5 (45) 33-42, 1951. As was pointed out by E. Ya. Remez, in certain cases the algorithm of M. V. Ostrogradski yields a better approximation than that given by a continued fraction. Unfortunately, up till now, the details of this theory, in particular the application for computational purposes, have not been fully worked out. (B.G.)

In a completely analogous way we can show that

$$\frac{a}{b} \leqq a_0+1.$$

Hence we can assume that $a_0 < a/b < a_0+1$. (In the cases that $a/b = a_0$ or $a/b = a_0+1$ the theorem would already have been proved, since $a/1 = p_0/q_0$ is a convergent and $(a_0+1)/1 = (p_0+p_{-1})/(q_0+q_{-1})$ is an intermediate fraction of the number α).

If the fraction a/b is not identical with a convergent or intermediate fraction of the number α, then it must lie between two consecutive intermediate fractions, i.e. for suitably chosen k and r ($k > 0$, $0 \leqq r < a_{k+1}$ or $k = 0$, $1 \leqq r < a_1$) it lies between the fractions

$$\frac{p_k r+p_{k-1}}{q_k r+q_{k-1}} \quad \text{and} \quad \frac{p_k(r+1)+p_{k-1}}{q_k(r+1)+q_{k-1}}$$

from which

$$\left|\frac{a}{b}-\frac{p_k r+p_{k-1}}{q_k r+q_{k-1}}\right| < \left|\frac{p_k(r+1)+p_{k-1}}{q_k(r+1)+q_{k-1}}-\frac{p_k r+p_{k-1}}{q_k r+q_{k-1}}\right|$$

$$= \frac{1}{\{q_k(r+1)+q_{k-1}\}\{q_k r+q_{k-1}\}}.$$

On the other hand it is clear that

$$\left|\frac{a}{b}-\frac{p_k r+p_{k-1}}{q_k r+q_{k-1}}\right| = \frac{m}{b(q_k r+q_{k-1})},$$

where m is a positive integer and as such at least equal to 1. Consequently

$$\frac{1}{b(q_k r+q_{k-1})} < \frac{1}{\{q_k(r+1)+q_{k-1}\}\{q_k r+q_{k-1}\}}$$

whence

$$q_k(r+1)+q_{k-1} < b.$$

The fraction

$$\frac{p_k(r+1)+p_{k-1}}{q_k(r+1)+q_{k-1}} \tag{25}$$

whose denominator is smaller than that of the fraction a/b thus lies nearer to the number α than the fraction

$$\frac{p_k r+p_{k-1}}{q_k r+q_{k-1}} \tag{26}$$

(since, from the results of § 4; it follows that for every successive intermediate fraction is closer to α than its predecessor). It must therefore lie closer to α than the fraction a/b which itself is enclosed between (25) and (26). This, however, contradicts the definition of a best approximation. Thus Theorem 15 has been proved.

When defining the concept of best approximation which was fundamental to our theorem, we estimated the degree of the approximation of the number α by the fraction $\frac{a}{b}$ by establishing how large the absolute value of the difference $\alpha - \frac{a}{b}$ was. This method of procedure seems to be the most natural. Nevertheless in the Theory of Numbers it is frequently more important or more convenient to consider, for this purpose, the difference $b\alpha - a$, which only differs from the previous difference by a factor of b, and the (absolute) magnitude of which can just as well serve to indicate how far the fraction a/b differs from α. This transposition from one criterium to another might at first sight appear to be trivial. And in many cases it is. However, we shall immediately discover that this is not always the case. This is because the factor b, which distinguishes the two differences, does not represent a constant quantity but depends upon the approximating fractions and changes with them.

From now on we shall refer to the best approximation as it was understood in Theorem 15, as *best approximation of the first kind*. Furthermore we shall call the rational fraction a/b $(b > 0)$ as the *best approximation of the second kind* to the number α if from $c/d \neq a/b$, $0 < d \leqq b$ there follows

$$|d\alpha - c| > |b\alpha - a|.$$

Thus best approximation of the second kind is determined by the magnitude of $|b\alpha - a|$, just as best approximation of the first kind was determined by the magnitude of $\left|\alpha - \frac{a}{b}\right|$.

It is easy to show that every best approximation of the second kind is at the same time a best approximation of the first kind.

If

$$\left|\alpha - \frac{c}{d}\right| \leqq \left|\alpha - \frac{a}{b}\right|, \quad \left(\frac{c}{d} \neq \frac{a}{b}, \quad d \leqq b\right)$$

holds, then by multiplying the terms of the left hand side of the inequality by d and those of the right by b, we obtain

$$|d\alpha - c| \leqq |b\alpha - a|.$$

Thus if the fraction a/b were not a best approximation of the first kind, then it could not be a best approximation of the second. Thus our assertion is proved.

The converse of this assertion would, however, be false; a best approximation of the first kind need not necessarily be a best approximation of the second. Indeed it is easy to see, for example, that the fraction $\frac{1}{3}$ is a best approximation of the first kind to the number $\frac{1}{5}$, but it is not a best approximation of the second kind. This can be seen from the inequalities

$$|1 \cdot \tfrac{1}{5} - 0| < |3 \cdot \tfrac{1}{5} - 1|, \quad 1 < 3.$$

From our remarks and from Theorem 15 it follows that all best approximations of the second kind are either convergents or intermediate fractions. But we can (and upon this is based the rôle which best approximation of the second kind plays in the theory of continued fractions) derive a far sharper result.

THEOREM 16. *Every best approximation of the second kind is a convergent.*

PROOF. Let the fraction a/b be a best approximation of the second kind to the number

$$\alpha = [a_0; a_1, a_2, \ldots],$$

whose convergents we shall denote by p_k/q_k. If now $a/b < a_0$, then we have

$$|1.\alpha - a_0| < \left|\alpha - \frac{a}{b}\right| \leqq |b\alpha - a|, \quad 1 \leqq b$$

i.e. a/b would not be a best approximation of the second kind. Thus $a/b \geqq a_0$. But in this case the fraction a/b, if it is not to be identically equal to one of the convergents, must either lie between two convergents p_{k-1}/q_{k-1} and p_{k+1}/q_{k+1} which are either both of even order or both of odd order, or be greater than p_1/q_1. In the first case

$$\left|\frac{a}{b} - \frac{p_{k-1}}{q_{k-1}}\right| \geqq \frac{1}{bq_{k-1}}$$

and

$$\left|\frac{a}{b}-\frac{p_{k-1}}{q_{k-1}}\right|<\left|\frac{p_k}{q_k}-\frac{p_{k-1}}{q_{k-1}}\right|=\frac{1}{q_kq_{k-1}},$$

whence

$$b>q_k. \tag{27}$$

On the other hand

$$\left|\alpha-\frac{a}{b}\right|\geqq\left|\frac{p_{k+1}}{q_{k+1}}-\frac{a}{b}\right|\geqq\frac{1}{bq_{k+1}}$$

and consequently

$$|b\alpha-a|\geqq\frac{1}{q_{k+1}}$$

whilst

$$|q_k\alpha-p_k|\leqq\frac{1}{q_{k+1}}.$$

From this follows

$$|q_k\alpha-p_k|\leqq|b\alpha-a|. \tag{28}$$

Relationships (27) and (28) show that a/b is not a best approximation of the second kind.

In the second case (i.e. when $a/b>p_1/q_1$)

$$\left|\alpha-\frac{a}{b}\right|>\left|\frac{p_1}{q_1}-\frac{a}{b}\right|\geqq\frac{1}{bq_1},$$

from which

$$|b\alpha-a|>\frac{1}{q_1}=\frac{1}{a_1}$$

follows. On the other hand we clearly have

$$|1.\alpha-a_0|\leqq\frac{1}{a_1}$$

so that

$$|b\alpha-a|>|1.\alpha-a_0|,\ 1\leqq b.$$

However, this again contradicts our definition of the best approximation of the second kind. Thus Theorem 16 has been proved in its entirety.

Now we should like to ask if it is possible to reverse Theorems 15 and 16. Firstly, as one can easily see, Theorem 15 is irreversible; intermediate fractions are not necessarily best approximations of the first kind. For example it is easy to see that the fraction $\frac{1}{2}$ is an intermediate fraction of the number $\frac{4}{5}$, however, it is not a best approximation of the second kind, for

$$|\tfrac{4}{5}-\tfrac{1}{1}| < |\tfrac{4}{5}-\tfrac{1}{2}|, \quad 1 < 2.$$

Any number of such examples can be constructed, as the reader can confirm for himself without much trouble.

Theorem 16, on the other hand, can be almost completely reversed, a consequence of which is that it gains considerably in significance.

THEOREM 17. *Every convergent is a best approximation of the second kind. The only (trivial) exception is given by*

$$\alpha = a_0+\tfrac{1}{2}, \quad \frac{p_0}{q_0} = \frac{a_0}{1}.$$

PRELIMINARY REMARK. In the case $\alpha = a_0+\frac{1}{2}$, the fraction $p_0/q_0 = a_0/1$ is indeed not a best approximation of the second kind, for then

$$|1.\alpha-(a_0+1)| = 1|1.\alpha-a_0|.$$

PROOF. We consider the expression

$$|y\alpha-x|, \tag{29}$$

in which y takes the values $1, 2, \ldots, q_k$ whilst x can assume any positive integer value. We denote by y_0 that value of y for which, with suitably chosen x, the expression (29) assumes its smallest value (if there are a number of values of y having this property, then we choose y_0 to be the *smallest* of them). We denote by x_0 that value of x for which $|y_0\alpha-x|$ assumes its smallest value. It is easy to see that this value is *unique*. For if we had

$$\left|\alpha-\frac{x_0}{y_0}\right| = \left|\alpha-\frac{x_0'}{y_0}\right|, \quad (x_0 \neq x_0')$$

then clearly

$$\alpha = \frac{x_0+x_0'}{2y_0}.$$

Now we shall assert that this fraction is irreducible. If, namely, $x_0+x_0' = lp$, $2y_0 = lq$ $(l > 1)$, we obtain for the case $l > 2$

$$q < y_0, \quad \alpha = \frac{p}{q}, \quad |q\alpha - p| = 0$$

which contradicts the definition of y_0. If, however, $l = 2$ then $q = y_0$ and

$$|q\alpha - p| = |y_0\alpha - p| = 0 < |y_0\alpha - x_0|$$

which contradicts the definition of x_0.

Thus developing the rational number α as a continued fraction, we obtain

$$\alpha = \frac{p_n}{q_n}, \qquad p_n = x_0+x_0', \qquad q_n = 2y_0 = a_n q_{n-1}+q_{n-2}, \qquad a_n \geqq 2.$$

Thus if either $a_n > 2$ or $a_n = 2$ when $n > 1$, then we have $q_{n-1} < y_0$, but

$$|q_{n-1}\alpha - p_{n-1}| = \frac{1}{q_n} = \frac{1}{2y_0} \leqq \tfrac{1}{2} \leqq |y_0\alpha - x_0|.$$

This again contradicts the definition of y_0. For $n = 1$, $a_n = 2$, however, we have $\alpha = a_0+\frac{1}{2}$, $y_0 = 1$ and we obtain the very case that was excluded.

The given conditions, then, serve uniquely to determine the values of y_0 and x_0. From this it follows immediately that x_0/y_0 is a best approximation of the second kind to the number α; for if the inequality

$$|b\alpha - a| \leqq |y_0\alpha - x_0| \qquad \left(\frac{a}{b} \neq \frac{x_0}{y_0}, \; b \leqq y_0\right)$$

holds, then our definitions of x_0 and y_0 would clearly by contradicted. Thus from Theorem 16 we have

$$x_0 = p_s, y_0 = q_s. \qquad (s \leqq k)$$

If $s = k$, the theorem is proved. If, however, $s < k$, then we have

$$|q_s\alpha - p_s| > \frac{1}{q_s+q_{s-1}} \geqq \frac{1}{q_{k-1}+q_k},$$

$$|q_k\alpha - p_k| \leqq q_{k+1}.$$

Since, by definition, $p_s = x_0$ and $q_s = y_0$

$$|q_s\alpha - p_s| \leqq |q_k\alpha - p_k|$$

then

$$\frac{1}{q_{k-1}+q_k} < \frac{1}{q_{k+1}}$$

i.e.

$$q_{k+1} < q_k + q_{k-1}.$$

But from the law of formation of the q_k this is impossible. Thus Theorem 17 has been proved.

From a historical point of view, the properties of continued fractions which we have established in this section provided the first motive for the discovery and study of the calculus of continued fractions. Huygens, who set himself the task of constructing a model of the solar system in the form of a system of gear wheels, had so to determine the number of teeth that their ratio for two coupled wheels (being equal to the ratio of the times of rotation of the two wheels) was as near as possible to the ratio α of the times of rotation of the corresponding planets. At the same time, for technical reasons, the numbers of teeth should not be excessively large. Thus a rational fraction must be determined whose numerators and denominator must not exceed a certain limit and at the same time must deviate as little as possible from the given number α (in theory this number can also be irrational, however, in practice such a number would be represented by a rational fraction with very large numerator and denominator). We have already seen that the theory of continued fractions offers the possibility of a complete solution to this problem.

§ 7 The Order of the Approximation

In the previous section we compared the magnitude of the difference $\left|\alpha - \frac{p_k}{q_k}\right|$ with others of the same form. Here we shall estimate the magnitude of the difference itself, as opposed to comparing it with others of its type. In order to be able to estimate how small $\left|\alpha - \frac{p_k}{q_k}\right|$ is, we proceed quite naturally by comparing it with some available function of q_k. In this sense Theorem 9 of

the previous section immediately yields the inequality [1])

$$\left|\alpha - \frac{p_k}{q_k}\right| < \frac{1}{q_k^2}. \tag{30}$$

The question now arises as to whether this inequality can be sharpened, i.e. if the right hand side may be replaced by some other function $f(q_k)$ of the denominator q_k, which satisfies the inequality

$$f(n) < \frac{1}{n^2}$$

for all $n \geqq 1$.

If we wish to establish that the sharpened inequality (30) should be satisfied for arbitrary α and any k, we soon discover that such a sharpening is impossible. In other words, however, small ε is taken to be we can always construct a case for which the inequality

$$\left|\alpha - \frac{p_k}{q_k}\right| > \frac{1-\varepsilon}{q_k^2}$$

holds. To convince ourselves of this it suffices to consider the number

$$\alpha = [0; n, 1, n] = \frac{n+1}{n(n+2)}$$

for which

$$p_1 = 1, q_1 = n, p_3 = n+1, q_3 = n(n+2),$$

and hence

$$\left|\alpha - \frac{p_1}{q_1}\right| = \left|\frac{p_3}{q_3} - \frac{p_1}{q_1}\right| = \frac{1}{n(n+2)} = \frac{1}{\left(q_1^2\left(1+\frac{2}{n}\right)\right)}.$$

Thus it is sufficient to choose such an n that the inequality

$$\frac{1}{1+\frac{2}{n}} > 1-\varepsilon$$

[1]) If $\alpha = p_k/q_k$ (in which case Theorem 9 is inapplicable due to the non-existence of q_{k+1}) the inequality (30) becomes trivial.

should hold; we then obtain

$$\left|\alpha-\frac{p_1}{q_1}\right|>\frac{1-\varepsilon}{q_1^2}.$$

If, however, we dispense with the requirement that the sharpened inequality should be satisfied for any α and for *all* values of k *without exception*, then we obtain, as we shall shortly see, a series of interesting and important results.

THEOREM 18. *If the number α possesses a convergent of order $k>0$, then at least one of the two inequalities*

$$\left|\alpha-\frac{p_k}{q_k}\right|<\frac{1}{2q_k^2},$$

$$\left|\alpha-\frac{p_{k-1}}{q_{k-1}}\right|<\frac{1}{2q_{k-1}^2},$$

holds.

PROOF. Since α lies between the two convergents p_{k-1}/q_{k-1} and p_k/q_k, we have

$$\left|\alpha-\frac{p_k}{q_k}\right|+\left|\alpha-\frac{p_{k-1}}{q_{k-1}}\right|=\left|\frac{p_k}{q_k}-\frac{p_{k-1}}{q_{k-1}}\right|=\frac{1}{q_kq_{k+1}}<\frac{1}{2q_k^2}+\frac{1}{2q_{k-1}^2}$$

(the last inequality merely expresses the fact that the geometric mean of the quantities $1/q_k^2$ and $1/q_{k-1}^2$ is less than their arithmetic mean; the sign of equality would only be possible for $q_k=q_{k-1}$). From this the assertion of our theorem follows directly.

The theorem which has just been proved is of particular interest in that it is possible, in a certain sense, to reverse it.

THEOREM 19. *Every irreducible fraction a/b satisfying the inequality*

$$\left|\alpha-\frac{a}{b}\right|<\frac{1}{2b^2}$$

is a convergent of the number α.

PROOF. From Theorem 16 it suffices to prove that the fraction a/b is a best approximation of the second kind for the number α. Let

$$|d\alpha-c|\leqq|b\alpha-a|<\frac{1}{2b}\qquad\left(d>0,\ \frac{c}{d}\neq\frac{a}{b}\right),$$

then

$$\left|\alpha-\frac{c}{d}\right|<\frac{1}{2bd}$$

and consequently

$$\left|\frac{c}{d}-\frac{a}{b}\right|\leqq\left|\alpha-\frac{c}{d}\right|+\left|\alpha-\frac{a}{b}\right|<\frac{1}{2bd}+\frac{1}{2b^2}=\frac{b+d}{2b^2d}. \tag{31}$$

On the other hand, since $cd\neq a/b$, then

$$\left|\frac{c}{d}-\frac{a}{b}\right|\geqq\frac{1}{bd}$$

from which the inequality (31) becomes

$$\frac{1}{bd}<\frac{b+d}{2b^2d}$$

from which, in turn, $d>b$. Thus the fraction a/b is indeed a best approximation of the second kind to the number α, and thus Theorem 19 has been proved.

A further sharpening of Theorem 18 is given by the following, much deeper

THEOREM 20 [1]). *If the number α possesses a convergent of order $k>1$, then at least one of the three inequalities*

$$\left|\alpha-\frac{p_k}{q_k}\right|<\frac{1}{\sqrt{5}q_k^2},$$

$$\left|\alpha-\frac{p_{k-1}}{q_{k-1}}\right|<\frac{1}{\sqrt{5}q_{k-1}^2},$$

$$\left|\alpha-\frac{p_{k-2}}{q_{k-2}}\right|<\frac{1}{\sqrt{5}q_{k-2}^2}$$

is satisfied.

PROOF. We put for $k\geqq 1$

$$\frac{q_{k-2}}{q_{k-1}}=\phi_k,\quad \phi_k+r_k=\psi_k.$$

[1]) A somewhat simplified version of the proof given here is to be found in the paper by I. I. Zhogin, "A variation of the proof of a theorem in the theory of continued fractions", Advances in Mathematical Science, 12, 3, 321-322, 1957. (B.G.)

LEMMA. *If for* $k \geqq 2$, $\psi_k \leqq \sqrt{5}$ *and* $\psi_{k-1} \leqq \sqrt{5}$, *then*

$$\phi_k > \frac{\sqrt{5}-1}{2}.$$

Indeed, since

$$\frac{1}{\phi_{n+1}} = \frac{q_n}{q_{n-1}} = a_n + \phi_n$$

and

$$r_n = a_n + \frac{1}{r_{n+1}}$$

then

$$\frac{1}{\phi_{n+1}} + \frac{1}{r_{n+1}} = \phi_n + r_n = \psi_n.$$

From the conditions of the lemma, however,

$$\phi_k + r_k \leqq \sqrt{5}, \quad \frac{1}{\phi_k} + \frac{1}{r_k} \leqq \sqrt{5},$$

whence

$$(\sqrt{5}-\phi_k)\left(\sqrt{5}-\frac{1}{\phi_k}\right) \geqq 1$$

or, since ϕ_k is a rational number,

$$5-\sqrt{5}\left(\phi_k - \frac{1}{\phi_k}\right) > 0.$$

From this, bearing in mind that $\phi_k > 0$

$$\left(\frac{\sqrt{5}}{2} - \phi_k\right)^2 < \tfrac{1}{4}$$

and consequently

$$\frac{\sqrt{5}}{2} - \phi_k < \tfrac{1}{2},$$

$$\phi_k > \frac{\sqrt{5}-1}{2}$$

and thus the lemma has been proved.

Now let us assume that, contrary to our assertion,

$$\left|\alpha - \frac{p_n}{q_n}\right| \geqq \frac{1}{\sqrt{5}q_n^2} \quad (n = k,\ k-1,\ k-2).$$

From formula (16) of Chapter I we have

$$\left|\alpha-\frac{p_n}{q_n}\right| = \left|\frac{p_n r_{n+1}+p_{n-1}}{q_n r_{n+1}+q_{n-1}}-\frac{p_n}{q_n}\right|$$

$$= \frac{1}{q_n(q_n r_{n+1}+q_{n-1})} = \frac{1}{q_n^2(r_{n+1}+\phi_n)} = \frac{1}{q_n^2\psi_{n+1}}$$

and thus

$$\psi_{n+1} \leqq \sqrt{5} \quad (n = k, k-1, k-2).$$

From the lemma we obtain

$$\phi_k > \frac{\sqrt{5}-1}{2}, \quad \phi_{k+1} > \frac{\sqrt{5}-1}{2},$$

and hence from inequality (32)

$$\alpha_k = \frac{1}{\phi_{k+1}} - \phi_k < \frac{2}{\sqrt{5}-1} - \frac{\sqrt{5}-1}{2} = 1$$

which is clearly impossible. This contradiction yields the proof of Theorem 20.

Theorems 18 and 20 might lead one to suppose that they constitute the beginning of a whole sequence of theorems having a similar form. This is, however, false. To establish this, we consider the number

$$\alpha[1; 1, 1, \ldots].$$

Writing, as usual, $\alpha = 1+\frac{1}{r_1}$, we clearly obtain $r_1 = \alpha$, from which

$$\alpha = 1+\frac{1}{\alpha}, \quad \alpha^2-\alpha-1 = 0$$

and hence

$$\alpha = \frac{1+\sqrt{5}}{2}.$$

Since $r_n = \alpha$ for any n, then

$$\alpha = \frac{p_k\alpha+p_{k-1}}{q_k\alpha+q_{k-1}}$$

and consequently

$$\left|\alpha-\frac{p_k}{q_k}\right| = \frac{1}{q_k(q_k\alpha+q_{k-1})} = \frac{1}{q_k^2\left(\alpha+\frac{q_{k-1}}{q_k}\right)}.$$

From Theorem 6 of Chapter I, however,

$$\frac{q_k}{q_{k-1}} = [1; 1, 1, \ldots, 1] \to \alpha. \quad (k \to \infty)$$

From this follows

$$\frac{q_{k-1}}{q_k} = \frac{1}{\alpha} + \varepsilon_k = \frac{\sqrt{5}-1}{2} + \varepsilon_k \quad (\varepsilon_k \to 0 \text{ for } k \to \infty).$$

Thus

$$\left|\alpha - \frac{p_k}{q_k}\right| = \frac{1}{q_k^2\left(\frac{\sqrt{5}+1}{2} + \frac{\sqrt{5}-1}{2} + \varepsilon_k\right)} = \frac{1}{q_k^2(\sqrt{5}+\varepsilon_k)}.$$

This shows that for any choice of $c < 1/\sqrt{5}$ and for sufficiently large k, we must have

$$\left|\alpha - \frac{p_k}{q_k}\right| > \frac{c}{q_k^2}.$$

Thus if the corresponding inequality is to be satisfied for an infinite number of values of k and for any α, the constant $1/\sqrt{5}$ of Theorem 20 cannot be replaced by any smaller constant. For any smaller constant we can always find some α $\left(\text{for example, } \alpha = \frac{\sqrt{5}+1}{2}\right)$ for which the corresponding inequality is only satisfied for a finite number of values of k. In particular the sequence of Theorems which began with Theorems 18 and 20 terminates with Theorem 20; it cannot be continued.

§ 8 General Laws of Approximation

Up till now we have only been interested in the sort of approximation which is given by the convergents. In this way we were able to solve a series of problems which were connected with this question. Since we have established that the degree of approximation provided by the convergents is, in a certain sense, the best, we should expect that the results derived would allow us to investigate the laws which govern the approximation of irrational numbers by means of rational fractions. At the same time we can

expect that no dependence on any specific representational apparatus will arise. To this complex of questions we shall now turn our attention. Clearly we cannot, within the framework of our elementary introduction, give a complete treatment of the fundamentals of the theory in question. This is due not so much to shortage of space as to the fact that these matters have very little immediate relationship to our problem. Clearly we shall have to limit ourselves to giving a series of elementary results which illustrate the application of continued fractions to the examination of the arithmetic nature of irrational numbers.

The first point of connection with the problems of the previous section can be formulated in the following way: for which constants c does the inequality

$$\left|\alpha - \frac{p}{q}\right| < \frac{c}{q^2} \tag{33}$$

possess, for arbitrary real α, infinitely many solutions in integers p and q $(q > 0)$? The concluding result of the last section leads us easily to the following

THEOREM 21. *For arbitrary real α the inequality* (33) *has infinitely many solutions in integers p and q $(q > 0)$, if $c \geqq 1/\sqrt{5}$. If, however, $c < 1/\sqrt{5}$, then it is possible to find such an α that the inequality* (33) *has no more than a finite number of such solutions.*

In point of fact our first assertion is an immediate consequence of Theorem 20 (in the case that $\alpha(= a/b)$ is rational and consequently possesses but a finite number of convergents, the first assertion of Theorem 20 permits of a trivial proof, for one puts $q = nb$, $p = na$, $n = 1, 2, \ldots$).

Therefore let $c < 1/\sqrt{5}$. As in § 7 we put

$$\alpha = \frac{1+\sqrt{5}}{2} = [1; 1, 1, \ldots].$$

We know from Theorem 19 that if the integers p and q $(q > 0)$ satisfy the inequality (33), then p/q is a convergent of the number α. At the end of § 7 we saw, however, that of these convergents at most a finite number satisfy the inequality (33) if, as assumed, $c < 1/\sqrt{5}$. Hence our assertion has been proved in its entirety.

Thus in general, taking all possible real α into account, the degree of approximation characterised by the quantity $1/(\sqrt{5}q^2)$ cannot be

improved upon. This does not, however, mean that in the case of certain irrational numbers a far higher degree of approximation is impossible. On the contrary, the possibilities for this to occur are without limit; we can show this most easily with the help of the continued fraction itself.

THEOREM 22. *However the positive function $\phi(q)$ of the positive integer q may behave, it is always possible to find an irrational number α such that the inequality*

$$\left|\alpha - \frac{p}{q}\right| < \phi(q)$$

should possess infinitely many solutions in integers p and q $(q > 0)$.

PROOF. We construct the infinite continued fraction α by choosing its elements to be such that they satisfy the inequality

$$a_{k+1} > \frac{1}{q_k^2 \phi(q_k)}, \qquad (k \geqq 0)$$

which clearly can be brought about in an infinity of ways (a_0 can be chosen arbitrarily). For any $k \geqq 0$ we have

$$\left|\alpha - \frac{p_k}{q_k}\right| < \frac{1}{q_k q_{k+1}} = \frac{1}{q_k(a_{k+1}q_k + q_{k-1})} \leqq \frac{1}{q_{k+1}q_k^2} < \phi(q_k)$$

and thus our theorem has been proved.

We wish to point out that in the most general case the inequalities

$$\frac{1}{q_k(q_k + q_{k+1})} < \left|\alpha - \frac{p_k}{q_k}\right| \leqq \frac{1}{q_k q_{k+1}},$$

or equivalently

$$\frac{1}{q_k^2\left(a_{k+1} + 1 + \dfrac{q_{k-1}}{q_k}\right)} < \left|\alpha - \frac{p_k}{q_k}\right| \leqq \frac{1}{\left(q_k^2 a_{k+1} + \dfrac{q_{k-1}}{q_k}\right)},$$

yield the inequalities

$$\frac{1}{q_k^2(a_{k+1} + 2)} < \left|\alpha - \frac{p_k}{q_k}\right| \leqq \frac{1}{q_k^2 a_{k+1}}. \qquad (34)$$

It can be seen from these that for a given $a_0, a_1, \ldots, a_k$, the larger the next element a_{k+1} is, the better the fraction p_k/q_k approximates

α. Since, in any case, the convergents provide the best approximation, we conclude that those irrational numbers are best approximated by rational fractions whose elements contain large numbers. This assertion, which is of a qualitative nature, finds its quantitative expression in the inequality (34). In particular those irrational numbers are worst approximated whose elements are bounded. Now it becomes clear why we repeatedly chose the number

$$\frac{\sqrt{5}+1}{2} = [1; 1, 1, \ldots]$$

when we required an irrational number which would not permit of a better degree of approximation than that of a given order. Of all the irrational numbers it is the one with the smallest possible elements (in the case that a_0 is not taken into account, which is of no significance here) and hence allows the worst possible degree of approximation by rational fractions.

The specific characteristics of the sort of approximation which is displayed by numbers with bounded elements are reflected completely in the following theorem, which in view of our previous reasoning appears to be almost trivial.

THEOREM 23. *For every irrational number α with bounded elements, the inequality*

$$\left|\alpha - \frac{p}{q}\right| < \frac{c}{q^2} \tag{35}$$

has, for sufficiently small c, no solution in integers p and q $(q > 0)$. Conversely, for every number α with unbounded elements, the inequality (33) *is satisfied for arbitrary $c > 0$, by infinitely many integers p and q.*

In other words irrational numbers with bounded elements cannot be approximated to a higher order than $1/q^2$. As opposed to this every irrational number with unbounded elements can be better approximated.

PROOF. If among the elements of the continued fraction which represents α there exist any number of large ones, we can, for arbitrary $c > 0$, find an infinite number of values of k for which

$$a_{k+1} > \frac{1}{c}.$$

Hence, from the second of the inequalities (34)

$$\left|\alpha-\frac{p_k}{q_k}\right|<\frac{c}{q_k^2}$$

for infinitely many k, whence the second assertion of our theorem has been proved.

If, however, an $M>0$ exists for which

$$a_k<M \quad (k=1,2,\ldots)$$

then we obtain from the first of the inequalities (34)

$$\left|\alpha-\frac{p_k}{q_k}\right|>\frac{1}{q_k^2(M+2)}$$

for any $k\geqq 0$.

From this it follows, for an arbitrary pair of integers p and q, that if one determines the index k by means of the inequality

$$q_{k-1}<q\leqq q_k$$

and bears in mind that all convergents are best approximations of the first kind, then

$$\begin{aligned}\left|\alpha-\frac{p}{q}\right|\geqq\left|\alpha-\frac{p_k}{q_k}\right|&>\frac{1}{q_k^2(M+2)}\\ &=\frac{1}{q^2(M+2)}\left(\frac{q}{q_k}\right)^2>\frac{1}{q^2(M+2)}\left(\frac{q_{k-1}}{q_k}\right)^2\\ &=\frac{1}{q^2(M+2)}\left(\frac{q_{k-1}}{a_kq_{k-1}+q_{k-2}}\right)^2\\ &>\frac{1}{q^2(M+2)}\,\frac{1}{(q_k+1)^2}>\frac{1}{(M+2)(M+1)^2q^2}.\end{aligned}$$

Now choosing

$$c<\frac{1}{(M+2)(M+1)^2}$$

the inequality (33) cannot be satisfied by any integers p and q. Thus the first assertion of our theorem is proved.

Up till now we have always assessed the degree of approximation by means of the magnitude of the difference $\alpha-\frac{p}{q}$. Just as in § 6

we could instead of this have estimated the difference $q\alpha - p$ in all the theorems that were proved, and introduced suitable modifications in all their formulations. This simple remark reads at once to an extremely important point of view in the problem under consideration.

The simplest homogeneous linear equation in the unknown x and y

$$\alpha x - y = 0$$

where α is a given irrational number, clearly cannot be solved exactly in terms of whole numbers [1]). Nevertheless we can speak of an approximate solution to this problem, i.e. a choice of integers p and q such that the difference $\alpha x - y$ does not exceed a certain quantity. Clearly all the previous results of this paragraph may serve as assertions concerning the laws by which approximate integer solutions to equation (35) exist. For example Theorem 21 shows that there always exist infinitely many integers x and y $(x > 0)$ such that

$$|\alpha x - y| < \frac{C}{x} \tag{36}$$

if C represents a positive number which is not smaller than $1/\sqrt{5}$.

From this new point of view it would seem completely natural to proceed from the homogeneous equation (35) to the inhomogeneous equation

$$\alpha x - y = \beta \tag{37}$$

(in which β is a prescribed arbitrary number), and to inquire into the possibility and indeed the kinds of solution which it possesses in integers x and y. One has, that is, to inquire into the problems that arise in an attempt to render the difference $\alpha x - y - \beta$ as small as possible by appropriate choice of the integers x and y.

This problem was first proposed by the great Russian mathematician P. L. Tschebyscheff, who established the first important results which are connected with it. At the present time this problem is the subject of intensive study, in which the Soviet arithmetic school actively participates.

The first important detail which distinguishes the inhomogeneous from the homogeneous case, consists in that the number α must be

[1]) Apart from the trivial solution $x = y = 0$.

irrational if it is to be possible, for arbitrary β, and by a suitable choice of the integers x and y, to render the quantity $|\alpha x-y-\beta|$ arbitrarily small. (In the homogeneous case, on the contrary, this can be achieved for the quantity $|\alpha x-y|$ for *arbitrary* α).

Indeed, if $\alpha = a/b$ with integer $(b > 0)$ and α, and we put $\beta = 1/2b$, then we obtain for arbitrary integer values of x and y

$$|\alpha x-y-\beta| = \left|\frac{2(ax-by)-1}{2b}\right| \geqq \frac{1}{2b},$$

since $|2(ax-by)-1|$, an odd number, is at least equal to unity.

We shall therefore assume in our further deliberations that the number α is irrational. Subject to this condition we shall see that not only can the quantity $|\alpha x-y-\beta|$ be rendered arbitrarily small, but that the analogy with the homogeneous case goes far deeper.

THEOREM 24. (Tschebyscheff). *For arbitrary irrational* α *and for arbitrary real* β *the inequality*

$$|\alpha x-y-\beta| < \frac{3}{x}$$

has infinitely many solutions in integers x *and* y [1]).

PRELIMINARY REMARK. Clearly this result corresponds completely to that property of homogeneous equations which was brought to expression in Theorem 21. The only difference consists in that the constant $1/\sqrt{5}$ has been replaced by the number 3. The order of approximation remains, however, the same. It should be pointed out that the number 3 is certainly not the smallest possible: the lower bound of the numbers which may replace it without affecting the validity of the theorem, is considerably smaller.

PROOF. Let p/q be an arbitrary convergent of the number α. Then

$$\alpha = \frac{p}{q} + \frac{\delta}{q^2} \qquad (0 < \delta < 1). \tag{38}$$

For any choice of the real number β an integer t can always be found

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[1]) A simpler proof of a somewhat stronger theorem was given by A. Ya. Khintchine in the paper "Dirichlet's Principle in the theory of Diophantine Approximation", Advances in Mathematical Science, part 3, 1948 (see pages 17-18). A further refinement was contained in the article by A. Ya. Khintchine "On a problem of Tschebyscheff", Proc. Acad. Sci. U.S.S.R., Math. Ser., 10, 281-294, 1946. (B.G.)

such that

$$|q\beta - t| \leqq \tfrac{1}{2},$$

whence

$$\beta = \frac{t}{q} + \frac{\delta'}{2q} \qquad (|\delta'| \leqq 1). \tag{39}$$

Since p and q are coprime, a pair of integers x and y exist which satisfy the condition [1])

$$\frac{q}{2} \leqq x < \frac{3q}{2}, \quad px - qy = t.$$

In this case, however, we have from (38) and (39)

$$|\alpha x - y - \beta| = \left| \frac{xp}{q} + \frac{x\delta}{q^2} - y - \frac{t}{q} - \frac{\delta'}{2q} \right|$$

$$= \left| \frac{x\delta}{q^2} - \frac{\delta'}{2q} \right| < \frac{x}{q^2} + \frac{1}{2q},$$

and since

$$q > \tfrac{2}{3}x$$

then

$$|\alpha x - y - \beta| < \frac{9}{4x} + \frac{3}{4x} = \frac{3}{x}.$$

Finally, the number q can be chosen to be arbitrarily large. But since $x \geq q/2$, then x is also arbitrarily large; clearly from this our theorem has been proved.

The problem of solving equation (37) approximately by means of integers can, however, be posed in another way which may even perhaps appear to be a little more natural. Since the essence of the problem consists of making the quantity $|\alpha x - y - \beta|$ as small as possible by choosing x and y to be as small as possible it would

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[1]) Here is the proof: If r/s is that convergent of the number α which immediately precedes p/q, then

$$qr - ps = \varepsilon = \pm 1, \; q(\varepsilon rt) - p(\varepsilon st) = t\varepsilon^2 = t$$

and for any integer value of k

$$p(kq - \varepsilon st) - q(kp - \varepsilon rt) = t$$

k may, however, be so chosen that

$$\frac{q}{2} \leqq x = kq - \varepsilon st < \frac{3q}{2}.$$

appear to be most natural to formulate the problem in the following way: If an arbitrary large integer n is given then we know (from Theorem 24 which we have just proved) that for any irrational number α and for any real β, integers $x > 0$ and y can be found such that the inequality

$$|\alpha x - y - \beta| < \frac{1}{n} \tag{40}$$

is satisfied. However, Theorem 24 does not, in general, tell us between what bounds these numbers must be looked for in order that the accuracy indicated by the number $1/n$ can be attained. This object could be made capable of achievement if, for example, one could prescribe a number N dependent indeed upon n but not upon α and β, with the property that the inequality (40) should be satisfied with the supplementary condition $|x| \leqq N$.

Quite clearly this new sort of problem differs very greatly from those with which we have previously been concerned. Where previously the accuracy of the approximation was determined by its dependence on the magnitude of the number x, now we prescribe the accuracy and ask how large x must be chosen such that this accuracy can be achieved. Corresponding to this distinction in the posing of the problem, we find that their solutions are of quite different kinds. In particular we obtain quite different results in the homogeneous and inhomogeneous cases.

In the case of the homogeneous equation ($\beta = 0$) our problem has a very simple solution.

THEOREM 25. *However the real numbers $n \geqq 1$ and α are constituted, integers x and y can always be found such that the unequalities*

$$0 < x \leqq n, \quad |\alpha x - y| < \frac{1}{n}$$

are satisfied.

PROOF. If α is rational, $\alpha = a/b$ and $0 < b \leqq n$, thus the proof of the theorem is rendered trivial by putting $x = b$ and $y = a$. However, if α cannot be expressed in this way, i.e. if α is irrational or is a rational fraction whose numerator exceeds n, then, determining the index k from the relationship

$$q_k \leqq n \leqq q_{k+1}$$

(here p_k/q_k denotes a convergent of the number α), we obtain

$$\left|\alpha-\frac{p_k}{q_k}\right| \leqq \frac{1}{q_k q_{k+1}} < \frac{1}{q_k n},$$

and consequently

$$|\alpha q_k - p_k| < \frac{1}{n},$$

whence our theorem has been proved.

Now we quite naturally proceed to inquire as to whether the same degree of approximation is encountered in the case of the inhomogeneous equation, i.e. as to whether it can be stated that corresponding to any arbitrary irrational number α a positive number C can be found such that how ever the numbers $n \geqq 1$ and β should be chosen, integers x and y exist which satisfy the following inequalities:

$$0 < x \leqq Cn, \quad |\alpha x - y - \beta| < \frac{1}{n}.$$

(Evidently we require less here than we proved for the homogeneous case; since we allow C to depend on α, whilst in the homogeneous case $C = 1$ is taken to be an absolute constant.) After a little reflection it is not difficult to see that our supposition is improbable. Firstly it is false for rational α; since in this case we have seen that in general (i.e. for any β) the quantity $|\alpha x - y - \beta|$ cannot be made arbitrarily small. This fact would lead us to expect that for some α which, although irrational, is nevertheless very near to some rational number, relatively large values of x and y are required to make the quantity $|\alpha x - y - \beta|$ small (though by Theorem 24, this can be done). The same considerations lead us to presume further that the difference between $\alpha x - y$ and an arbitrary real number β can be the more easily approximated the worse the number α can be approximated by a rational fraction, i.e. the more difficult it is to cause the quantity $\alpha x - y$ to be nearly zero. For this, as we already know, the elements of the number must not increase too rapidly.

All these considerations are confirmed, and given quantitative expression, by the following theorem.

THEOREM 26. *If a positive number C is to exist, with the property that for any real numbers $n \geqq 1$ and β, integers x and y can be found $(x > 0)$ such that the inequalities*

$$x \leqq Cn, \quad |\alpha x - y - \beta| < \frac{1}{n}$$

are satisfied, then it is necessary and sufficient that the irrational number α should be represented by a continued fraction with bounded elements.

PROOF. Let

$$\alpha = [a_0; a_1, a_2, \ldots]$$

and $a_i < M$ $(i = 1, 2, \ldots)$. Further let $m \geqq 1$, and β be an arbitrary real number. We denote the convergents of the number α by p_k/q_k and determine the index k from the inequality

$$q_k \leqq m < q_{k+1}.$$

Then

$$\left|\alpha - \frac{p_k}{q_k}\right| < \frac{1}{q_k q_{k+1}} < \frac{1}{mq_k}$$

i.e.

$$\alpha = \frac{p_k}{q_k} + \frac{\delta}{mq_k} \quad (|\delta| \leqq 1). \tag{41}$$

Further we choose the integer t to be such that

$$|\beta q_k - t| \leqq \tfrac{1}{2},$$

whence

$$\beta = \frac{t}{q_k} + \frac{\delta'}{2q_k} \quad (|\delta'| \leqq 1). \tag{42}$$

Finally we determine, just as was done in the proof of Theorem 24, a pair of integers x and y satisfying the relationship

$$xp_k - yq_k = t, \quad 0 < x \leqq q_k. \tag{43}$$

From (41), (42) and (43) there follows

$$\begin{aligned} |\alpha x - y - \beta| &= \left|\frac{xp_k}{q_k} - y - \frac{t}{q_k} + \frac{x\delta}{mq_k} - \frac{\delta'}{2q_k}\right| \\ &= \left|\frac{x\delta}{mq_k} - \frac{\delta'}{2q_k}\right| < \frac{x}{mq_k} + \frac{1}{2q_k} \leqq \frac{1}{m} + \frac{1}{2q_{k+1}}\left(\frac{q_{k+1}}{q_k}\right) \\ &< \frac{1}{m} + \frac{1}{2m}(a_{k+1}+1) < \frac{1}{m} + \frac{M+1}{2m} = \frac{M+3}{2m}. \end{aligned}$$

Up till now the number $m \geqq 1$ was chosen quite arbitrarily. Now, $n \geqq 1$ being given, we put $m = \dfrac{M+3}{2} n$; clearly we have $m > 1$ and from the above

$$0 < x \leqq q_k \leqq m = \frac{M+3}{2} n,$$

$$|\alpha x - y - \beta| < \frac{1}{n},$$

if the numbers x and y are chosen in the prescribed manner. Thus the first part of the theorem has been proved.

For the proof of the second part we shall assume that arbitrarily large quantities occur among the elements of the number α. In this case we know from Theorem 23 that however small the number ε may be chosen, integers $q > 0$ and p may be found, such that the inequality

$$\left| \alpha - \frac{p}{q} \right| < \frac{\varepsilon^2}{q^2}$$

holds. From this

$$\alpha = \frac{p}{q} + \frac{\delta \varepsilon^2}{q^2} \qquad (|\delta| < 1).$$

Now we put $n = q/\varepsilon$ and $\beta = 1/2q$. Then we obtain for any integers x and y $(0 < x \leqq Cn)$

$$|\alpha x - y - \beta| = \left| \frac{xp}{q} - y - \frac{1}{2q} + \frac{x\delta\varepsilon^2}{q^2} \right| = \left| \frac{2(xp-yq)-1}{2q} + \frac{x\delta\varepsilon^2}{q^2} \right|$$

$$> \frac{|2(xp-yq)-1|}{2q} - \frac{x\varepsilon^2}{q^2} \geqq \frac{1}{2q} - \frac{C\varepsilon}{q} = \frac{1-2C\varepsilon}{2q} = \frac{1-2C\varepsilon}{2\varepsilon} \cdot \frac{1}{n}.$$

However great C may be, for sufficiently small ε we obtain $\dfrac{1-2C\varepsilon}{2\varepsilon} > 1$, and consequently for any integer values of x and y $(0 < x \leqq Cn)$

$$|\alpha x - y - \beta| > \frac{1}{n}.$$

Thus the second part of the theorem has been proved.

We wish now to assemble the results which we have derived. When examining the approximate solutions to equation (37) in integers we must consider as the normal case that in which, for any $n \geqq 1$ and a certain $x < Cn$, the degree of accuracy characterised by the quantity $1/n$ can be achieved. In this C is a constant which can depend on α. The homogeneous (i.e. that resulting from $\beta = 0$) equation always possesses a normal solution (Theorem 25). Theorem 26 shows us that the general (inhomogeneous) equation admits of a normal solution if, and only if, the corresponding homogeneous equation possesses no "supra-normal" solution, i.e. if for any $\varepsilon > 0$ and suitably chosen n it can be satisfied (to within a discrepancy of $1/n$) by integers $x > 0$ and y, where $x < \varepsilon n$. Considered from this point of view the results of our inquiry can be regarded as a special case of the general rule for the solution of linear equations (algebraic equations, integral equations and so on): *In the general case an inhomogeneous equation admits of a normal solution if the corresponding homogeneous equation does not admit of a supranormal solution.*

It remains to be mentioned that in Theorem 26 we made the quantity C independent of β. One can, without destroying the result, permit C to be a function of β. But the proof of the second part would then have been considerably more difficult.

§ 9 The Approximation of Algebraic Numbers. Liouville's Transcendental Numbers

Let

$$f(x) = a_0 + a_1 x + \ldots + a_n x^n \tag{44}$$

be a polynomial of the n-th degree with *integer* coefficients a_0, $a_1, \ldots, a_n$. A number α which is a root of such a polynomial is called an *algebraic* number. Since every rational number ($\alpha = a/b$) can be defined as the root of an equation of the first degree ($bx - a = 0$) the concept of the algebraic number is clearly a generalisation of the concept of the rational number. If a given algebraic number satisfies an equation of the n-th degree $f(x) = 0$ and does not satisfy an equation of lesser degree (with integer coefficients) then it is called an *algebraic number of the n-th degree*. In particular rational numbers may be defined as algebraic numbers of the first degree. The number $\sqrt{2}$, being a root of the polynomial $x^2 = 2$ is an

algebraic number of the second degree, or as one says, a *quadratic irrational*. Cubic irrationals, irrationals of the fourth degree, and so on, are defined in the same way. All non-algebraic numbers are called *transcendental* numbers. To these belong, for example, e and π. By reason of the great part that algebraic numbers play in modern Number Theory, many special investigations have been devoted to their properties in respect of their approximation by rational fractions. The first significant result in this direction was the following so-called Theorem of Liouville.

THEOREM 27. *Corresponding to every algebraic number of the n-th degree there exists a positive number C such that for any integers p and q* $(q > 0)$

$$\left|\alpha - \frac{p}{q}\right| > \frac{C}{q^n}.$$

PROOF. Let α be a root of the polynomial (44). As we know from Algebra, we can write

$$f(x) = (x-\alpha)f_1(x), \tag{45}$$

where $f_1(x)$ is a polynomial of degree $n-1$. It is easy to see that $f_1(\alpha) \neq 0$. Indeed if the equation $f_1(\alpha) = 0$ were to be satisfied, then $f_1(x)$ would be divisible by $(x-\alpha)$, i.e. $(x-\alpha)^2$ would be a factor of $f(x)$. In this case, however, $(x-\alpha)$ would be a factor of $f'(x)$, i.e. $f'(\alpha) = 0$. This is impossible since $f'(x)$ is a polynomial of the $(n-1)$-th degree with integer coefficients and α is an algebraic number of the n-th degree. Thus

$$f_1(\alpha) \neq 0.$$

Consequently a positive number δ can be found such that

$$f_1(x) \neq 0. \qquad (\alpha-\delta \leqq x \leqq \alpha+\delta)$$

Let p and q $(q > 0)$ be any pair of integers. If $|\alpha-p/q| \leqq \delta$ then $f_1(p/q) \neq 0$. Putting $x = p/q$ in the identity (45), we obtain

$$\frac{p}{q} - \alpha = \frac{f\left(\frac{p}{q}\right)}{f_1\left(\frac{p}{q}\right)} = \frac{a_0 + a_1\left(\frac{p}{q}\right) + \ldots + a_n\left(\frac{p}{q}\right)^n}{f_1\left(\frac{p}{q}\right)}$$

$$= \frac{a_0 q^n + a_1 p q^{n-1} + \ldots + a_n p^n}{q^n f_1\left(\frac{p}{q}\right)}.$$

The numerator of the last fraction is a non-zero integer, since otherwise $\alpha = p/q$, and from the conditions of the theorem α is an irrational number. Consequently the absolute value of this number is at least 1. Now denote by M the upper bound of the function $f_1(x)$ in the interval $(\alpha-\delta, \alpha+\delta)$; then we obtain from the last equation

$$\left|\alpha - \frac{p}{q}\right| \geqq \frac{1}{Mq^n}.$$

In the case that

$$\left|\alpha - \frac{p}{q}\right| > \delta$$

then certainly

$$\left|\alpha - \frac{p}{q}\right| > \frac{\delta}{q^n}.$$

Denoting by C an arbitrary positive number which is smaller than both δ and $1/M$, then in all cases (i.e. for any integers $q > 0$ and p)

$$\left|\alpha - \frac{p}{q}\right| > \frac{C}{q^n};$$

thus Theorem 27 has been proved.

The theorem of Liouville clearly implies that algebraic numbers cannot be approximated by means of rational fractions to an accuracy which exceeds a certain order. This depends essentially on the degree of the algebraic number. The chief historical significance of this result was that it enabled the existence of transcendental numbers to be proved, and examples of such numbers to be constructed. To do this, it suffices, as we can see, to construct an irrational number that can be extremely well approximated by rational numbers. In this respect, as we know from Theorem 22, the possibilities are limitless.

If for any $C > 0$ and for any positive integer n, integers p and $q(q > 0)$ exist such that the inequality

$$\left|\alpha - \frac{p}{q}\right| \leqq \frac{C}{q^n} \tag{46}$$

is satisfied, then Theorem 27 shows that the number α is transcendental. Any amount of such numbers can be constructed with

the help of continued fractions. After the elements $a_1, a_2, \ldots, a_k$ have been chosen, it suffices to construct the convergents p_k/q_k and to make

$$a_{k+1} > q_k^{k-1}.$$

Then

$$\left|\alpha - \frac{p_k}{q_k}\right| < \frac{1}{q_k q_{k+1}} < \frac{1}{q_k^2 a_{k+1}} < \frac{1}{q_k^{k+1}}$$

from which it follows that the inequality (46) is satisfied for all sufficiently large values of k, however, the constant $C > 0$ and the positive integer n may be chosen.

§ 10 Quadratic Irrationals and Periodic Continued Fractions

Theorem 27 shows that a positive number C (independent of α) exists for which the inequality

$$\left|\alpha - \frac{p}{q}\right| < \frac{C}{q^2}$$

can have no solution in integral p and $q(q > 0)$. From Theorem 23 it follows that the elements of any quadratic irrational are bounded. But long before Liouville, Lagrange had discovered a much more profound property of the continued fractions which represent quadratic irrationals. It transpires that the sequence of elements of a quadratic irrational is always periodic and that conversely every periodic continued fraction represents a quadratic irrational. The present section is devoted to a proof of this theorem.

We shall call the continued fraction

$$\alpha = [a_0; a_1, a_2, \ldots]$$

periodic, if integers k_0 and h exist such that for any $k \geqq k_0$

$$a_{k+h} = a_k.$$

As in the case of a decimal fraction we shall write such a continued fraction

$$\alpha = [a_0; a_1, a_2, \ldots, a_{k_0-1}, \overline{a_{k_0}, a_{k_0+1}, \ldots, a_{k_0+h-1}}] \qquad (47)$$

THEOREM 28. *Every periodic continued fraction represents a quadratic irrational. Conversely every quadratic irrational can be represented by means of a periodic continued fraction.*

PROOF. The first statement can be proved simply. In the event the remainders of a periodic continued fraction satisfy the condition

$$r_{k+h} = r_k. \qquad (k \geqq k_0)$$

Thus from formula (16) of Chapter I we have for $k \geqq k_0$

$$\alpha = \frac{p_{k-1}r_k + p_{k-2}}{q_{k-1}r_k + q_{k-2}} = \frac{p_{k+h-1}r_{k+h} + p_{k+h-2}}{q_{k+h-1}r_{k+h} + q_{k+h-2}}$$

$$= \frac{p_{k+h-1}r_k + p_{k+h-2}}{q_{k+h-1}r_k + q_{k+h-2}}, \qquad (48)$$

from which follows

$$\frac{p_{k-1}r_k + p_{k-2}}{q_{k-1}r_k + q_{k-2}} = \frac{p_{k+h-1}r_k + p_{k+h-2}}{q_{k+h-1}r_k + q_{k+h-2}}.$$

Thus the number r_k satisfies a quadratic equation with integer coefficients and is therefore a quadratic irrational. However, in this case the first of equations (48) shows that α must also be a quadratic irrational.

The converse of this assertion is not dismissed so lightly. Suppose that the number α satisfies the quadratic equation

$$a\alpha^2 + b\alpha + c = 0 \qquad (49)$$

with integer coefficients. Substituting the expression

$$\alpha = \frac{p_{n-1}r_n + p_{n-2}}{q_{n-1}r_n + q_{n-2}}$$

(which expresses α in terms of its remainder of n-th order) in place of α, we see that r_n satisfies the equation

$$A_n r_n^2 + B_n r_n + C_n = 0 \qquad (50)$$

in which A_n, B_n and C_n are integers determined by the formulae

$$\left.\begin{aligned} A_n &= ap_{n-1}^2 + bp_{n-1}q_{n-1} + cq_{n-1}^2 \\ B_n &= 2ap_{n-1}p_{n-2} + b(p_{n-1}q_{n-2} + p_{n-2}q_{n-1}) + 2cq_{n-1}q_{n-2} \\ C_n &= ap_{n-2}^2 + bp_{n-2}q_{n-2} + cp_{n-2}^2. \end{aligned}\right\} \qquad (51)$$

In particular there follows from these

$$C_n = A_{n-1}. \qquad (52)$$

It can easily be verified, with the help of these formulae, that

$$B_n^2-4A_nC_n = (b^2-4ac)(p_{n-1}q_{n-2}-q_{n-1}p_{n-2})^2 = b^2-4ac, \quad (53)$$

i.e. that the discriminant of equation (50) is the same for all n and equal to the discriminant of equation (49).

Since

$$\left|\alpha-\frac{p_{n-1}}{q_{n-1}}\right| < \frac{1}{q_{n-1}^2}$$

there follows

$$p_{n-1} = \alpha q_{n-1}+\frac{\delta_{n-1}}{q_{n-1}}. \qquad (|\delta_{n-1}| < 1)$$

Hence the first of formulae (51) yields

$$A_n = a\left(\alpha q_{n-1}+\frac{\delta_{n-1}}{q_{n-1}}\right)^2+b\left(\alpha q_{n-1}+\frac{\delta_{n-1}}{q_{n-1}}\right)q_{n-1}+cq_{n-1}^2$$

$$= (a\alpha^2+b\alpha+c)q_{n-1}^2+2a\alpha\delta_{n-1}+a\frac{\delta_{n-1}^2}{q_{n-1}^2}+b\delta_{n-1},$$

from which using equation (49)

$$|A_n| = \left|2a\alpha\delta_{n-1}+a\frac{\delta_{n-1}^2}{q_{n-1}^2}+b\delta_{n-1}\right| < 2|a\alpha|+|a|+|b|$$

and, taking note of equation (52),

$$|C_n| = |A_{n-1}| < 2|a\alpha|+|a|+|b|.$$

Thus the absolute magnitudes of the coefficients A_n and C_n of equation (50) are bounded, from which they can only take on a finite number of different values as n varies. From equation (53) the same is true of B_n.

As n increases from 1 to ∞ only a finite number of equations of the form (50) can arise. In this event r_n can only have a finite number of values, so for suitably chosen h and k

$$r_k = r_{k+h}.$$

This shows that the continued fraction representing α is periodic. Thus the second assertion of our theorem has been proved.

No properties of the representing continued fraction, analogous to those which have just been proved, are known for algebraic numbers of higher degree. Nearly everything that is known about

the rational fraction approximation of algebraic numbers of higher degree consists of elementary consequences of Liouville's Theorem and certain theorems of more recent date which are sharpened forms of Liouville's Theorem. It is of interest to point out that up till the present time no continued fraction development of an algebraic number of higher degree than the second is known. It is not even known if such a development has bounded elements, and so on. Generally speaking the problems associated with the continued fraction expansion of algebraic numbers of degree higher than the second are extremely difficult and virtually unstudied.

III. The Metric Theory of Continued Fractions

§ 11 Introduction

In the previous two chapters we saw that with respect to their arithmetic properties the real numbers can be of quite different kinds. Apart from the fundamental division of real numbers into rational and irrational (and these into algebraic and transcendental) numbers, a much finer subdivision, based upon a series of properties which characterise the arithmetic nature of these numbers, can be established. Most fundamentally they can be characterised by the degree to which they can be approximated by rational fractions. Up till now we have contented ourselves with establishing the fact that numbers with this or that arithmetic property actually exist. Thus we know that there are numbers that cannot be approximated by a rational fraction p/q to an order higher than $1/q^2$ (for example the quadratic irrationals). On the other hand we know that numbers exist which admit a higher degree of approximation (Theorem 22, Chapter II). The question naturally prompts itself as to which of these two opposing properties must be regarded as the more *general*, which of the two types occurs the *more frequently*.

If we wish to accord this problem an exact formulation we must recognise that every time we prescribe some arithmetic property of real numbers (such as irrationality or being transcendental or having bounded elements, and so on) we divide the totality of all real numbers into two sets. One is the set of numbers which display this property, the other is the set of numbers which do not.

The task which we have set ourselves clearly turns upon the problem of comparing these two sets and establishing which is the larger and which is the smaller. Sets of real numbers can, however, be regarded from different points of view and compared with one another in terms of various properties. One can ask after their power, their measure, and a variety of other characteristics. The most interesting, as much in regard to the methods employed as in the results obtained by its use, is the *metric theory*, which inquires

into the measure of the set of numbers which are characterised by a certain property. The resulting theory, which may be called the *metric arithmetic of the continuum* has been extensively developed in recent times, and has yielded many simple and interesting results. As in all investigations into the arithmetic nature of irrational numbers, the apparatus of the continued fraction is the best and most natural implement of research. If, however, this apparatus is to be exploited as a device of *metric* arithmetic, i.e. if it is to be used to determine the measure of the set of numbers characterised by a certain arithmetic property possessed by them, then before all else we must subject this apparatus itself to a thorough and many-sided metrical analysis. In other words we must learn how to determine *the measure of a set of numbers whose continued fraction representations display this or that prescribed property*. Such problems can be of quite different kinds. Thus we can inquire after the measure of the set of numbers for which $a_4 = 2$ or $q_{10} < 1000$ or which have a bounded sequence of elements or among whose elements no even integer is to be found, and so on. The totality of methods which serve for the solution of problems of this kind constitutes the *metric theory of continued fractions*, to the fundamentals and elementary applications of which this chapter is dedicated. Since the fundamental arithmetic properties of a number are not altered if an arbitrary integer is added to it, we shall consider in the following only those real numbers lying between 0 and 1 (i.e. we consistently assume that $a_0 = 0$). In the event this restriction to a finite interval is necessary for the metric theory, if we wish to ensure that in general the measures of the sets encountered are not to be infinite.

We shall assume on the part of the reader that he has at his disposal a knowledge of the elementary theory of the measure of linear sets [1]).

§ 12 The Elements as Functions of the Number Being Represented

Every real number can be developed *in only one way* as a continued fraction

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[1]) For an understanding of this chapter the contents of Chapter 6 of the textbook by P. S. Alexandroff and A. N. Kolmogorov on the Theory of Functions of a Real Variable (Moscow 1933) are more than sufficient.

$$\alpha = [a_0; a_1, a_2, \ldots].$$

Thus every element is determined uniquely when the number α is given, i.e. it is a single valued function of α:

$$a_n = a_n(\alpha).$$

When constructing the metric theory of continued fractions the first step must be to carry out such an inquiry into the nature of this function as will at least give us some idea of its general behaviour. This takes place in the present section.

As has already been stated in § 11 we assume throughout that $a_0 = 0$. To simplify the notation we shall write

$$\alpha = [a_1, a_2, \ldots]$$

in place of

$$\alpha = [a_0; a_1, a_2, \ldots].$$

Thus we put

$$[a_1, a_2, \ldots] = \cfrac{1}{a_1 + \cfrac{1}{a_2 + \ldots}}.$$

We now investigate the first element a_1 as a function of α. Since

$$\frac{1}{\alpha} = a_1 + \cfrac{1}{a_2 + \ldots}$$

then clearly $a_1 = [1/\alpha]$, i.e. a_1 is the smallest number not exceeding the fraction $1/\alpha$. Thus

$$a_1 = 1 \text{ for } 1 \leqq \frac{1}{\alpha} < 2, \text{ i.e. } \tfrac{1}{2} < \alpha \leqq 1,$$

$$a_1 = 2 \text{ for } 2 \leqq \frac{1}{\alpha} < 3, \text{ i.e. } \tfrac{1}{3} < \alpha \leqq \tfrac{1}{2},$$

$$a_1 = 3 \text{ for } 3 \leqq \frac{1}{\alpha} < 4, \text{ i.e. } \tfrac{1}{4} < \alpha \leqq \tfrac{1}{3}, \text{ and so on.}$$

In the general case

$$a_1 = k \text{ for } k \leqq \frac{1}{\alpha} < k+1, \text{ i.e. } \frac{1}{k+1} < \alpha \leqq \frac{1}{k}.$$

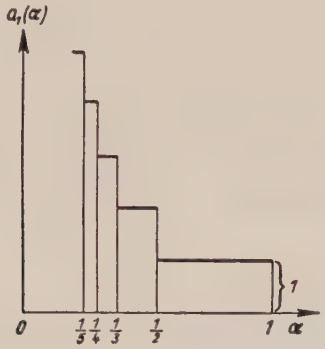


Fig. 1

Thus the function $a_1 = a_1(\alpha)$ has a point of discontinuity for all those values of α for which $1/\alpha$ is an integer; moreover as $\alpha \to 0$ it increases without limit. The function is shown in Fig. 1.

We wish further to point out that a_1 is constant in every interval $(1/(k+1) < \alpha \leqq 1/k)$. We shall call this interval *an interval of rank one*. Furthermore

$$\int_0^1 a_1(\alpha)d\alpha = +\infty$$

since the integral is clearly equivalent to the divergent series

$$\sum_{k=1}^{\infty} k\left(\frac{1}{k} - \frac{1}{k+1}\right) = \sum_{k=1}^{\infty} \frac{1}{k+1}.$$

Now let us proceed to investigate the function $a_2(\alpha)$. In order to do this we consider any fixed interval of rank one

$$\frac{1}{k+1} < \alpha \leqq \frac{1}{k}.$$

$a_1 = k$ throughout this interval, and consequently

$$\alpha = \cfrac{1\cdot}{k+\cfrac{1}{r_2}},$$

where $1 \leqq r_2 < \infty$ and $a_2 = [r_2]$. As r_2 increases from 1 to ∞, α

increases from $1/(k+1)$ to $1/k$, i.e. it runs through the prescribed interval of rank one. Thus, clearly.

$$a_2 = 1 \text{ for } 1 \leqq r_2 < 2, \text{ i.e. } \frac{1}{k+1} \leqq \alpha < \frac{1}{k+\frac{1}{2}},$$

$$a_2 = 2 \text{ for } 2 \leqq r_2 < 3, \text{ i.e. } \frac{1}{k+\frac{1}{2}} \leqq \alpha < \frac{1}{k+\frac{1}{3}},$$

$$a_2 = 3 \text{ for } 3 \leqq r_2 < 3, \text{ i.e. } \frac{1}{k+\frac{1}{3}} \leqq \alpha < \frac{1}{k+\frac{1}{4}},$$

and generally

$$a_2 = l \text{ for } l \leqq r_2 < l+1, \text{ i.e. } \frac{1}{k+\dfrac{1}{l}} \leqq \alpha < \frac{1}{k+\dfrac{1}{l+1}}.$$

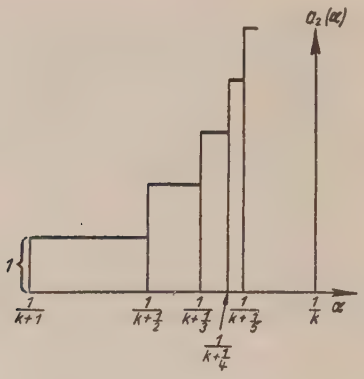


Fig. 2

Over the interval of rank one which we prescribed, the function $a_2(\alpha)$ has the form given in Fig. 2.

In each of the intervals

$$\left(\frac{1}{k+\dfrac{1}{l}}, \frac{1}{k+\dfrac{1}{l+1}}\right)$$

which we shall refer to as as *an interval of rank two*, the function $a_2(\alpha)$ is constant. Every interval of rank one can therefore be decomposed into a denumerable set of intervals of rank two which run from left to right (we recall that the intervals of rank one form

a sequence which progresses from right to left). The set of points for which $a_1 = k$ constitutes an interval of rank one; the set for which $a_2 = 1$ is a denumerable set of intervals of rank two (each of which belongs to a different interval of rank one). Every interval of rank one is determined by a condition of the form $a_1 = k$, every interval of rank two by conditions of the form $a_1 = k$ and $a_2 = 1$.

Now let us suppose that the intervals of ranks up to n have been defined and that the behaviour of the functions $a_1(\alpha)$, $a_2(\alpha)$, $\ldots$, $a_n(\alpha)$ has been investigated. Every system of values

$$a_1 = k_1, a_2 = k_2, \ldots, a_n = k_n \tag{54}$$

determines an interval J_n of rank n. In order to investigate the behaviour of the function $a_{n+1}(\alpha)$ in the interval J_n, we point out that any number belonging to this interval can be expressed in the form

$$\alpha = [k_1, k_2, \ldots, k_n, r_{n+1}]. \tag{55}$$

Here r_{n+1} takes all possible values from 1 to ∞. Conversely for any r_{n+1} $(1 < r_{n+1} < \infty)$ the expression (55) yields a number α for which the conditions (54) are obeyed and thus belongs to the interval J_n. Since, however, $a_{n+1} = [r_{n+1}]$ we see that in each interval of rank n, the function $a_{n+1}(\alpha)$ takes on all integer values between 1 and ∞. In order to obtain a clearer picture we shall, as usual, denote by p_k/q_k the convergents of the number α. Then

$$\alpha = \frac{p_n r_{n+1} + p_{n-1}}{q_n r_{n+1} + q_{n-1}}.$$

As α runs through the interval J_n in question, r_{n+1} increases from 1 to ∞, whilst p_n, q_n, p_{n-1} and q_{n-1} remain constant since they are completely determined by the numbers $a_1, a_2, \ldots, a_n$ which preserve their values in the interval J_n. Letting r_{n+1} take in turn the values $r_{n+1} = 1$ and $r_{n+1} = \infty$ we obtain as end points of the interval J_n the points

$$\frac{p_n + p_{n-1}}{q_n + q_{n-1}} \text{ and } \frac{p_n}{q_n}.$$

Since

$$\alpha - \frac{p_n}{q_n} = \frac{p_n r_{n+1} + p_{n-1}}{q_n r_{n+1} + q_{n-1}} - \frac{p_n}{q_n} = \frac{(-1)^n}{q_n(q_n r_{n+1} + q_{n-1})},$$

then α is a monotonic function of r_{n+1} in the interval $(1, \infty)$. In consequence we know that conversely r_{n+1} and hence a_{n+1} are monotonic functions of α in the interval

$$J_n = \left(\frac{p_n}{q_n}, \frac{p_n+p_{n-1}}{q_n+q_{n-1}}\right).$$

As α runs through the interval J_n, then $a_{n+1}(\alpha)$ takes in succession the values $1, 2, \ldots$ and partitions the interval J_n into a denumerably infinite number of intervals of rank $(n+1)$. This sequence runs from right to left for even n and from left to right for odd n.

In this way the structure of the function $a_n(\alpha)$ has been fully explained, at least from the qualitative side. We now agree to call the interval $(0, 1)$ (the unit interval) an interval of rank zero, and cover it with a mesh of increasingly small intervals, by embedding in every interval of rank n a sequence of intervals of rank $n+1$. This sequence runs from right to left for even n and from left to right for odd n. The function $a_{n+1}(\alpha)$ $(n = 0, 1, \ldots)$ is constant in each interval of rank $n+1$. It is monotonic and over each interval of rank n takes every value from 1 to ∞. To every system of values

$$a_1 = k_1, a_2 = k_2, \ldots, a_n = k_n$$

there corresponds a uniquely determined interval of rank n, and conversely. The more general system of values

$$a_{m_1} = k_1, a_{m_2} = k_2, \ldots, a_{m_s} = k_s$$

determines in general a denumerably infinite set of intervals.

The first problem in the metric theory of continued fractions which suggests itself consists of the determination of the set of points of the interval $(0, 1)$ for which $a_n = k$. We already know that this set always forms a system of intervals. We have, therefore, to determine the total length of these intervals. A first approximation to the solution of this problem is easily obtained.

In the following we shall agree to denote by

$$E\begin{pmatrix} n_1, n_2, \ldots, n_s \\ k_1, k_2, \ldots, k_s \end{pmatrix}$$

the set of points in the interval $(0, 1)$ for which the conditions

$$a_{n_1} = k_1, a_{n_2} = k_2, \ldots, a_{n_s} = k_s$$

are fulfilled. Here all n_i and k_i are clearly integers; the n_i are all distinct. We already know that such a set represents a system of intervals. In particular the set

$$E\begin{pmatrix}1, & 2, & \ldots, & n\\ k_1, & k_2, & \ldots, & k_n\end{pmatrix}$$

as we already know, is an interval of rank n which is characterised by the relationship

$$a_i = k_i. \qquad (i = 1, 2, \ldots, n)$$

Clearly we always have

$$\sum_{k_l=1}^{\infty} E\begin{pmatrix}n_1, \ldots, n_{l-1}, n_l, n_{l+1}, \ldots, n_s\\ k_1, \ldots, k_{l-1}, k_l, k_{l+1}, \ldots k_s\end{pmatrix} = E\begin{pmatrix}n_1, \ldots, n_{l-1}, n_{l+1}, \ldots, n_s\\ k_1, \ldots, k_{l-1}, k_{l+1}, \ldots, k_s\end{pmatrix}. \qquad (56)$$

Lastly we agree to denote by $\mathfrak{M}E$ the measure of the set E.

Now we consider an arbitrary interval

$$J_n = E\begin{pmatrix}1, & 2, & \ldots, & n\\ k_1, & k_2, & \ldots, & k_n\end{pmatrix}$$

of rank n, and an interval

$$J_{n+1}^{(s)} = E\begin{pmatrix}1, & 2, & \ldots, & n, & n+1\\ k_1, & k_2, & \ldots, & k_n, & s\end{pmatrix}$$

of rank $n+1$ contained in it. We know already that the end points of J_n are

$$\frac{p_n}{q_n} \text{ and } \frac{p_n+p_{n-1}}{q_n+q_{n-1}},$$

where p_k/q_k denotes the general convergent of order k of the continued fraction

$$[k_1, k_2, \ldots, k_n].$$

On the other hand, for all points of the interval $J_{n+1}^{(s)}$,

$$a_{n+1} = [r_{n+1}] = s,$$

from which

$$s \leqq r_{n+1} < s+1.$$

Thus among those points

$$\alpha = \frac{p_n r_{n+1} + p_{n-1}}{q_n r_{n+1} + q_{n-1}}$$

of the interval J_n belong those points of the interval $J_{n+1}^{(s)}$ for which $s \leqq r_{n+1} < s+1$; i.e. the points

$$\frac{p_n s + p_{n-1}}{q_n s + q_{n-1}} \text{ and } \frac{p_n(s+1) + p_{n-1}}{q_n(s+1) + q_{n-1}}$$

are the end points of the interval $J_{n+1}^{(s)}$.

From this follows

$$\mathfrak{M} J_n = \left| \frac{p_n}{q_n} - \frac{p_n + p_{n-1}}{q_n + q_{n-1}} \right| = \frac{1}{q_n(q_n + q_{n-1})} = \frac{1}{q_n^2 \left(1 + \frac{q_{n-1}}{q_n}\right)},$$

$$\mathfrak{M} J_{n+1}^{(s)} = \left| \frac{p_n s + p_{n-1}}{q_n s + q_{n-1}} - \frac{p_n(s+1) + p_{n-1}}{q_n(s+1) + q_{n-1}} \right|$$

$$= \frac{1}{[q_n s + q_{n-1}][q_n(s+1) + q_{n-1}]}$$

$$= \frac{1}{q_n^2 s^2 \left(1 + \frac{q_{n-1}}{q_n}\right)\left(1 + \frac{1}{s} + \frac{q_{n-1}}{s q_n}\right)},$$

and hence

$$\frac{\mathfrak{M} J_{n+1}^{(s)}}{\mathfrak{M} J_n} = \frac{1}{s} \cdot \frac{1 + \frac{q_{n-1}}{q_n}}{\left(1 + \frac{q_{n-1}}{s q_n}\right)\left(1 + \frac{1}{s} + \frac{q_{n-1}}{s q_n}\right)}.$$

Here the second factor on the right hand side is clearly always less than 2 and greater than $\frac{1}{3}$ $\left(\text{this last due to } \frac{1 + \frac{q_{n-1}}{q_n}}{1 + \frac{q_{n-1}}{s q_n}} \geqq 1 \text{ and } 1 + \frac{1}{s} + \frac{q_{n-1}}{s q_n} < 3\right)$. Hence we obtain

$$\frac{1}{3s^2} < \frac{\mathfrak{M} J_{n+1}^{(s)}}{\mathfrak{M} J_n} < \frac{2}{s^2}. \tag{57}$$

This shows that in any arbitrary interval of rank n, that interval of rank $n+1$ which is characterised by the value $a_{n+1} = s$ takes up a part of the order $1/s^2$. It is of the greatest importance that the bounds given in the inequality (57) are independent both of the numbers $k_1, k_2, \ldots, k_n$ and of the rank n, and are determined solely by the number s. Writing the inequality in the form

$$\frac{\mathfrak{M}J_n}{3s^2} < \mathfrak{M}J_{n+1}^{(s)} < \frac{2\mathfrak{M}J_n}{s^2},$$

summing over all intervals of rank n (or, equivalently, over $k_1, k_2, \ldots, k_n$ from 1 to ∞) and noting finally that clearly

$$\sum \mathfrak{M}J_n = 1 \text{ and } \sum \mathfrak{M}J_{n+1}^{(s)} = \mathfrak{M}E\binom{n+1}{s},$$

we obtain

$$\frac{1}{3s^2} < \mathfrak{M}E\binom{n+1}{s} < \frac{2}{s^2}.$$

Thus a first approximation to the solution of our problem has been obtained. We see that *the measure of the set of points for which a certain element has a given value s, always lies between* $1/3s^2$ *and* $2/s^2$ (*it is, in consequence, a magnitude of the order* $1/s^2$).

§ 13 The Metric Approximation of the Rate of Growth of the Elements

We already have at our disposal sufficient means for the solution of problems concerned with the measure of a set whose definition involves an infinite number of elements. As the first example of such a problem we prove the following simple

THEOREM 29. *The set of all numbers in the interval* (0, 1) *having bounded elements is of measure zero.*

PROOF. We denote by E_M the set of numbers in the interval (0, 1) whose elements are less than M. Let J_n be some interval of rank n the points of which satisfy the condition

$$a_i < M. \qquad (i = 1, 2, \ldots, n) \qquad (58)$$

The points in the interval J_n satisfying the supplementary condition $a_{n+1} = k$ form an interval of rank $n+1$ which we shall denote by $J_{n+1}^{(k)}$. From the first of the inequalities (57)

$$\mathfrak{M} J_{n+1}^{(k)} > \frac{1}{3k^2} \mathfrak{M} J_n,$$

whence

$$\mathfrak{M} \sum_{k \geqq M} J_{n+1}^{(k)} > \tfrac{1}{3} \mathfrak{M} J_n \sum_{k \geqq M} \frac{1}{k^2}$$

$$> \frac{1}{3} \mathfrak{M} J_n \sum_{i=1}^{\infty} \frac{1}{(M+i)^2} > \frac{1}{3} \mathfrak{M} J_n \int_{M+1}^{\infty} \frac{du}{u} = \frac{1}{3(M+1)} \mathfrak{M} J_n,$$

and since

$$\sum_{k=1}^{\infty} J_{n+1}^{(k)} = J_n$$

we also have

$$\mathfrak{M} \sum_{k < M} J_{n+1}^{(k)} > \left\{ 1 - \frac{1}{3(M+1)} \right\} \mathfrak{M} J_n = \tau \mathfrak{M} J_n, \tag{59}$$

where

$$\tau = 1 - \frac{1}{3(M+1)}.$$

Clearly $\tau < 1$ for $M > 0$.

Denoting by $E_M^{(n)}$ the set of those numbers of the interval (0, 1) which are characterised by the condition (58), we see from the inequality (59) that the part of the set $E_M^{(n+1)}$ which is enclosed in a certain interval J_n of rank n, is of smaller measure than $\tau \mathfrak{M} J_n$. Since, quite clearly, an interval of rank n which does not belong to the set $E_M^{(n)}$ (i.e. which does not satisfy the inequality (58)) cannot contain any point of the set $E_M^{(n+1)}$, then we have, by summing the inequality over all the intervals of rank n which belong to the set $E_M^{(n)}$

$$\mathfrak{M} E_M^{(n+1)} < \tau \mathfrak{M} E_M^{(n)}. \tag{60}$$

Successive application of this inequality clearly yields

$$\mathfrak{M} E_M^{(n+1)} < \tau^n \mathfrak{M} E_M^{(1)} \qquad (n \geqq 1)$$

whence, since $\tau < 1$, there follows

$$\mathfrak{M} E_M^{(n)} \to 0. \qquad (n \to \infty)$$

The set E_M, as defined above, is clearly contained in each of the sets $E_M^{(n)}$, whence

$$\mathfrak{M} E_M = 0$$

Putting further

$$\sum_{M=1}^{\infty} E_M = E,$$

we obtain

$$\mathfrak{M} E \leqq \sum_{M=1}^{\infty} \mathfrak{M} E_M = 0.$$

Clearly by making M sufficiently large, the set E_M may be made to contain every number with bounded elements, and thus our theorem has been proved.

We know already (from Theorem 23, Chapter II) that numbers α with bounded elements do not permit of a better degree of approximation by means of rational fractions than that indicated by the rule

$$\left| \alpha - \frac{p}{q} \right| < \frac{C}{q^2} \tag{61}$$

(to such numbers belong, for example, all quadratic irrational numbers). We now see that all these numbers constitute a set of measure zero. In other words, *almost all* numbers (i.e. all numbers apart from those of a set of measure zero) admit of a better degree of approximation by means of rational fractions. Evidently the main problem of metric approximation theory consists of the problem of determining the measure of the set of numbers which can be approximated by rational fractions to a prescribed degree. In particular one asks what is the best law of approximation which is obeyed by almost all (i.e. by all apart from a set of measure zero) numbers, i.e. within what bounds the law determined by the inequality (61) can be improved if one agrees that sets of numbers α of measure zero can be ignored.

We treat the solution of this problem in the next section. Here we prove the following

THEOREM 30. *Let $\phi(n)$ be an arbitrary positive function of the positive integer n. If the series $\sum_{n=1}^{\infty} 1/\phi(n)$ diverges, then the inequality*

$$a_n = a_n(\alpha) \geqq \phi(n) \tag{62}$$

is satisfied an infinite number of times for almost all α. *On the other hand if the series* $\sum_{n=1}^{\infty} \frac{1}{\phi(n)}$ *converges, then the inequality* (62) *is satisfied at most a finite number of times for almost all* α.

PRELIMINARY REMARK. Putting, in particular, the function $\phi(n)$ equal to a constant positive number M, then one concludes from Theorem 30 that the set of numbers E_M referred to in the proof of Theorem 29 is of measure zero. Hence Theorem 29 may be considered as the simplest special case of Theorem 30.

PROOF. The first assertion is proved in exactly the same way as in Theorem 29. Let J_{m+n} be an interval of rank $m+n$ for all points of which the inequality

$$a_{m+i} < \phi(m+i) \quad (i = 1, 2, \ldots, n) \tag{63}$$

is satisfied (we do not impose any conditions upon the $a_1, a_2, \ldots a_n$). Retaining the notation introduced in the proof of Theorem 29 we find analogously to the inequality (59)

$$\mathfrak{M} \sum_{k<\phi(m+n+1)} J_{m+n+1}^{(k)} < \left\{1 - \frac{1}{3(1+\phi(m+n+1))}\right\} \mathfrak{M} J_{m+n}.$$

Summing this inequality over all intervals of rank $m+n$ which satisfy the condition (63), and denoting by $E_{m,n}$ the set of all numbers in the interval (0, 1) satisfying this condition, then we clearly obtain

$$\mathfrak{M} E_{m,n+1} < \left\{1 - \frac{1}{3(1+\phi(m+n+1))}\right\} \mathfrak{M} E_{m,n}.$$

If the series $\sum_{n=1}^{\infty} 1/\phi(n)$ diverges, then clearly for any m the series

$$\sum_{i=2}^{\infty} \frac{1}{3(1+\phi(m+i))}$$

also diverges. From this follows, as we know from the theory of infinite products, that

$$\prod_{i=2}^{n} \left\{1 - \frac{1}{3(1+\phi(m+i))}\right\}$$

tends to zero as $n \to \infty$. Hence for any m,

$$\mathfrak{M}E_{m,n} \to 0. \qquad (n \to \infty)$$

Every number α for which

$$a_{m+i} < \phi(m+i) \qquad (i = 1, 2, \ldots)$$

holds, clearly belongs to all the sets

$$E_{m,n}. \qquad (n = 1, 2, \ldots)$$

Hence the set of all such numbers, which we shall denote by E_m, is clearly of measure zero. Putting finally,

$$E_1 + E_2 + \ldots + E_m + \ldots = E,$$

we see that $\mathfrak{M}E = 0$. But every number α for which the inequality (62) is satisfied not more than a finite number of times clearly belongs, for sufficiently large m, to the set E_m and hence to the set E. Thus the first assertion of our theorem has been proved.

Now let the series $\sum_{n=1}^{\infty} 1/\phi(n)$ converge. Let J_n be an interval of rank n and $J_{n+1}^{(k)}$ be an interval of rank $n+1$ contained in J_n and determined by means of the supplementary condition $a_{n+1} = k$. From the second of the inequalities (57) we have

$$\mathfrak{M}J_{n+1}^{(k)} < \frac{2}{k^2}\,\mathfrak{M}J_n;$$

from this follows

$$\begin{aligned}
\mathfrak{M} \sum_{k \geqq \phi(n+1)} J_{n+1}^{(k)} &< 2\mathfrak{M}J_n \sum_{k \geqq \phi(n+1)} \frac{1}{k^2} \\
&\leqq 2\mathfrak{M}J_n \sum_{i=0}^{\infty} \frac{1}{\{\phi(n+1)+i\}^2} \\
&< 2\mathfrak{M}J_n \left\{ \frac{1}{\phi(n+1)} + \int_{\phi(n+1)}^{\infty} \frac{du}{u^2} \right\} = \frac{4\mathfrak{M}J_n}{\phi(n+1)}.
\end{aligned}$$

Denoting by F_n the set of numbers in the interval $(0, 1)$ for which $a_n \geqq \phi(n)$, and summing the inequality obtained over all intervals J_n of rank n, then we have

$$\mathfrak{M}F_{n+1} < \frac{4}{\phi(n+1)},$$

since $\sum \mathfrak{M}J_n = 1$. Thus the measures of the sets $F_1, F_2, \ldots,$

$F_n, \ldots$ form a convergent series. Hence, denoting by F the set of all numbers in the interval $(0, 1)$ which belong to infinitely many sets F_n, then we obtain [1])

$$\mathfrak{M}F = 0.$$

But the set F is evidently just the set of numbers for which the inequality (62) is satisfied an infinite number of times. Thus the second statement of our theorem has been proved.

§ 14 The Metric Estimation of the Rate of Growth of the Denominators of the Convergents

The Main Theorem of the Metric Approximation Theory

THEOREM 31. *There exists a positive absolute constant B such that for sufficiently large n the inequality*

$$q_n = q_n(\alpha) < e^{Bn}$$

holds almost everywhere.

PRELIMINARY REMARK. In Chapter I, § 4 (Theorem 12) we saw that for all numbers α the denominator q_n cannot increase with n more slowly than a certain geometric progression with an absolute constant ratio. Theorem 31 shows that for almost all numbers α the numbers q_n do not grow more rapidly than another geometric progression whose ratio is also an absolute constant. This matter can also be expressed as follows: There exist two absolute constants a and A $(1 < a < A)$ such that for sufficiently large n the inequality

$$a < \sqrt[n]{q_n} < A$$

holds for almost all numbers α in the interval $(0, 1)$.

Actually a far sharper theorem holds: There exists an absolute constant γ such that

$$\sqrt[n]{q_n} \to \gamma \qquad (n \to \infty)$$

almost everywhere. Nevertheless the proof of this theorem is considerably more difficult and requires for its execution very profound methods whose acquaintance we shall not make until §§ 15

[1]) This is a well known theorem of metric set theory. Here is its proof: Clearly, for any m, the set F is contained in the set $\Sigma_{n=m}^{\infty} F_n$ whose measure does not exceed $\Sigma_{n=m}^{\infty} \mathfrak{M}F_n$, and consequently, by taking m sufficiently large, can be made arbitrarily small.

and 16. Unfortunately it is not possible to include the proof within the framework of this small book [1]). Nevertheless the property of the numbers q_n mentioned in Theorem 31 is quite sufficient for our next main aim which is the proof of Theorem 32.

PROOF. Denote by $E_n(g)$ $(n > 0, g \geqq 1)$ the set of numbers in the interval (0, 1) for which

$$a_1 a_2 \ldots a_n \geqq g.$$

Clearly this set corresponds to a system of intervals of rank n. The length of any one of these intervals is, as we know from § 12,

$$\left| \frac{p_n}{q_n} - \frac{p_n + p_{n-1}}{q_n + q_{n-1}} \right| = \frac{1}{q_n(q_n + q_{n-1})} < \frac{1}{q_n^2} < \frac{1}{(a_1 a_2 \ldots a_n)^2},$$

since successive application of the obvious inequality

$$q_n > a_n q_{n-1}$$

yields the inequality

$$q_n > a_n a_{n-1} \ldots a_2 a_1.$$

Therefore

$$\mathfrak{M} E_n(g) < \sum_{a_1 a_2 \ldots a_n \geqq g} \frac{1}{a_n^2 a_{n-1}^2 \ldots a_2^2 a_1^2} \tag{64}$$

where the summation is taken over all combinations of the positive integers $a_1, a_2, \ldots, a_n$ which satisfy the inequality $a_1 a_2 \ldots a_n \geqq g$. In order to estimate this sum we note that

$$\prod_{i=1}^{n} \frac{1}{a_i^2} = \prod \left(1 - \frac{1}{a_i}\right) \frac{1}{a_i(a_i+1)} \leqq 2^n \prod_{i=1}^{n} \frac{1}{a_i(a_i+1)}$$

$$= 2^n \prod_{i=1}^{n} \int_{a_i}^{a_i+1} \frac{dx_i}{x_i^2} = 2^n \int_{a_1}^{a_1+1} \int_{a_2}^{a_2+1} \cdots \int_{a_n}^{a_n+1} \frac{dx_1 dx_2 \ldots dx_n}{x_1^2 x_2^2 \ldots x_n^2}$$

and consequently

$$\sum_{a_1, a_2, \ldots, a_n \geqq g} \left\{ \prod_{i=1}^{n} \frac{1}{a_i^2} \right\} \leqq 2^n J_n(g).$$

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[1]) The proof of this fact was obtained by A. Ya. Khintchine in 1935 (see „Zur metrischen Kettenbruch-theorie", Compositio Mathematica, vol 3, part 2, 1936, 275-285). Shortly afterwards the French mathematician P. Lévy obtained an explicit expression for the constant γ, namely $\ln \gamma = \frac{\pi^2}{12 \ln 2}$ (see P. Lévy, Théorie de l'addition des variables aléatoires, Paris, 1937, 320). (B.G.)

Here $J_n(g)$ is the n-fold integral

$$\int\int\dots\int \frac{dx_1\,dx_2\dots dx_n}{x_1^2\,x_2^2\dots x_n^2}$$

taken over the region

$$x_i \geqq 1 \quad (i = 1, 2, \dots, n),$$

$$x_1 x_2 \dots x_n \geqq g.$$

When $g = 1$ this region clearly becomes the region $1 < x_i < \infty$ $(i = 1, 2, \dots, n)$ and we obtain

$$J_n(g) = \left\{\int_1^\infty \frac{dx}{x^2}\right\}^n = 1 \qquad (g \leqq 1). \tag{65}$$

Now we show that for $g > 1$

$$J_n(g) = \frac{1}{g}\sum_{i=0}^{n-1}\frac{(\ln g)^i}{i!}. \tag{66}$$

Indeed when $n = 1$ this equation reduces to

$$\int_g^\infty \frac{dx}{x^2} = \frac{1}{g},$$

i.e. it is correct. Now we assume that it is valid for $n = k$, then we obtain

$$J_{k+1}(g) = \int_1^\infty \frac{dx_{k+1}}{x_{k+1}^2} J_k\left(\frac{g}{x_{k+1}}\right) = \frac{1}{g}\int_0^g J_k(u)du$$

$$= \frac{1}{g}\left\{\int_0^1 J_k(u)du + \int_1^g J_k(u)du\right\}.$$

Replacing $J_k(u)$ in the first integral by means of formula (65) and in the second by means of formula (66) which we have assumed to be valid for $n = k$, $g \geqq 1$, then we find that

$$J_{k+1}(g) = \frac{1}{g}\left\{1 + \sum_{i=0}^{k-1}\frac{(\ln g)^{i+1}}{(i+1)!}\right\} = \frac{1}{g}\sum_{i=0}^{k}\frac{(\ln g)i}{i!}$$

which was to be proved.

Thus

$$\mathfrak{M}E_n(g) < \frac{2^n}{g}\sum_{i=0}^{n-1}\frac{(\ln g)^i}{i!}.$$

Putting, in particular, $g = e^{An}$, where $A > 1$ is constant, then we obtain

$$\mathfrak{M}E_n(e^{An}) < e^{n(\ln 2-A)} \sum_{i=0}^{n-1} \frac{(An)^i}{i!}.$$

In this sum it is easy to see that every term is smaller than

$$\frac{(An)^n}{n!}.$$

Thus, applying Stirling's formula for estimating $n!$ and letting, in the following, C_1 and C_2 be two positive absolute constants, we obtain

$$\mathfrak{M}E_n(e^{An}) < e^{n(\ln 2-A)}\, n\, \frac{(An)^n}{n!}$$

$$< C_1 e^{n(\ln 2-A)}\, \frac{n(An)^n}{n^n e^{-n}\sqrt{n}} < C_2\sqrt{n}e^{-n(A-\ln A-\ln 2-1)}.$$

If however, A is sufficiently large, then

$$A - \ln A - \ln 2 - 1 > 0,$$

and consequently $\mathfrak{M}E_n(e^{An})$ is smaller than the n-th term of a certain convergent series. Since, from this, the series

$$\sum_{n=1}^{\infty} \mathfrak{M}E_n(e^{An})$$

converges, then every number in the interval $(0, 1)$, with the exception of a set of measure zero, is an element of at most finitely many sets $E_n(e^{An})$. This means, however, that for almost all numbers in the interval $(0, 1)$

$$a_1 a_2 \ldots a_n < e^{An}$$

for sufficiently large n. Since, however,

$$q_n = a_n q_{n-1} + q_{n-2} < 2a_n q_{n-1}$$

and consequently

$$q_n < 2^n a_n a_{n-1} \ldots a_2 a_1$$

then for sufficiently large n

$$q_n < 2^n e^{An} = e^{Bn}$$

(where $B = A + \ln 2$) almost everywhere. Thus Theorem 31 has been proved.

This result, which is also of interest in its own right, is at this moment of especial importance to us since with its help we can easily prove the main problem of metric approximation theory which we announced in the preceding section.

THEOREM 32. *Let $f(x)$ be a positive continuous function of the positive variable x such that $xf(x)$ is a non-increasing function. If the integral*

$$\int_0^\infty f(x)dx \tag{67}$$

diverges, then for almost all α the inequality

$$\left|\alpha - \frac{p}{q}\right| < \frac{f(q)}{q} \tag{68}$$

has infinitely many solutions in integers p and $q(q > 0)$. On the other hand if the integral (67) *converges then for almost all α the inequality* (68) *has at most a finite number of solutions in integers p and q $(q > 0)$.*

PRELIMINARY REMARK. From Theorem 32 the inequality

$$\left|\alpha - \frac{p}{q}\right| < \frac{1}{q^2 \ln q}$$

for example, has infinitely many solutions almost everywhere whilst on the other hand for any constant $\varepsilon > 0$ the inequality

$$\left|\alpha - \frac{p}{q}\right| < \frac{1}{q^2 \ln q^{1+\varepsilon}}$$

has almost everywhere no more than a finite number of solutions. This gives a provisional idea of how much the fundamental laws of approximation change if we wish to disregard a set of measure zero.

PROOF 1. Let the integral (67) diverge. We put

$$\phi(x) = e^{Bx} f(e^{Bx})$$

where B is the constant of Theorem 31. Then the integral

$$\int_a^B \phi(x)dx = \frac{1}{B}\int_{Ba}^{BA} f(u)du$$

where $A > a > 0$, grows without limit as $A \to \infty$. Since, however,

from the conditions of the theorem, the function $\phi(x)$ is non-increasing, then the series

$$\sum_{n=1}^{\infty} \phi(n)$$

diverges. From Theorem 30 we conclude that the inequality

$$a_{i+1} \geqq \frac{1}{\phi(i)}$$

is satisfied almost everywhere for infinitely many values of i. From the validity of this inequality, however, follows

$$\left| \alpha - \frac{p_i}{q_i} \right| \leqq \frac{1}{q_i q_{i+1}} \leqq \frac{1}{a_{i+1} q_i^2} \leqq \frac{\phi(i)}{q_i^2}. \tag{69}$$

From Theorem 31, for sufficiently large i,

$$q_i < e^{Bi}$$

almost everywhere, whence

$$i > \frac{\ln q_i}{B}.$$

Therefore the inequality (69) leads, almost everywhere, to the inequality

$$\left| \alpha - \frac{p_i}{q_i} \right| \leqq \phi \frac{\left(\frac{\ln q_i}{B}\right)}{q_i^2} = \frac{f(q_i)}{q_i}$$

for sufficiently large i. The last inequality is thus satisfied almost everywhere for infinitely many values of i, whence the first assertion of our theorem has been proved.

2. Now let us assume that the integral (68), and in consequence also the series

$$\sum_{n=1}^{\infty} f(n),$$

converges. We denote by E_n the set of numbers α of the interval $(0, 1)$ which, for suitably chosen integer k, satisfy the inequality

$$\left| \alpha - \frac{k}{n} \right| < \frac{f(n)}{n},$$

(evidently the set E_n consists simply of the intervals of length $\frac{2f(n)}{n}$ whose centres lie at the points

$$\frac{1}{n}, \frac{2}{n}, \ldots, \frac{n-1}{n},$$

together with the intervals $\left(0, \frac{f(n)}{n}\right)$ and $\left(1-\frac{f(n)}{n}, 1\right)$.

Now

$$\mathfrak{M} E_n \leqq 2f(n)$$

(the $<$ sign holds in the case $f(n) > \frac{1}{2}$). Hence the series

$$\sum_{n=1}^{\infty} \mathfrak{M} E_n$$

converges. From this we conclude (as we have done many times already) that almost all numbers α in the interval (0, 1) can belong to at most a finite number of sets E_n. This, however, clearly implies that for sufficiently large positive integer q and any integer p, almost all numbers α of the interval (0, 1) satisfy the inequality

$$\left| \alpha - \frac{p}{q} \right| \geqq \frac{f(q)}{q},$$

whence the second assertion of our theorem has been proved.

In the next section we shall make the acquaintance of a method which enables us to solve far deeper problems of the metric theory of continued fractions.

§ 15 Gauss' Problem and the Theorem of Kusmin

In this section we shall consider a problem which historically was the first problem of the metric theory of continued fractions. This problem, which was posed by Gauss, was not solved until 1928[1]).

As we usual, we put

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[1]) In the paper by R. O. Kusmin "On a Problem of Gauss", Reports of the Academy of Sciences (A), 1928, 375-380. Another solution was proposed in the article by P. Lévy "Sur les lois de probabilité dont dépendent les quotients complets et incomplets d'une fraction continue", Bull. Soc. Math., 57, 1929, 178-194. (B.G.)

$$\alpha = [0; a_1, a_2, \ldots, a_n, \ldots],$$
$$r_n = r_n(\alpha) = [a_n; a_{n+1}, a_{n+2}, \ldots],$$

and denote by $z_n = z_n(\alpha)$ the value of the continued fraction

$$[0; a_{n+1}, a_{n+2}, \ldots],$$

i.e. we put

$$z_n = r_n - a_n.$$

Clearly we always have

$$0 \leqq z_n < 1.$$

We denote by $m_n(x)$ the measure of the set of numbers α in the interval $(0, 1)$ for which

$$z_n(\alpha) < x.$$

In one of his letters to Laplace, Gauss stated that he had succeeded in proving that

$$\lim_{n\to\infty} m_n(x) = \frac{\ln(1+x)}{\ln 2}. \qquad (0 \leqq x \leqq 1)$$

In the same letter Gauss pointed out that it would be extremely desirable to obtain an estimate for the difference

$$m_n(x) - \frac{\ln(1+x)}{\ln 2} \qquad (70)$$

for large n, in which he, as he wrote, had been unsuccessful. Apparently Gauss' proof was never published, and other proofs of this theorem were not forthcoming until 1928. In that year there appeared a proof of the Gauss conjecture discovered by R. O. Kusmin. At the same time Kusmin found a very good estimate for the difference (70). The purpose of this section is to describe Kusmin's results, together with some generalisations of which we shall later have need [1]).

Gauss knew already that the sequence of functions

$$m_0(x), \; m_1(x), \; m_2(x), \ldots, m_n(x), \ldots$$

[1]) R. O. Kusmin, as did Gauss, formulated his results in the language of the calculus of probability, which clearly has no affect on their metric content.

satisfies the functional equation

$$m_{n+1}(x) = \sum_{k=1}^{\infty} \left\{ m_n\left(\frac{1}{k}\right) - m_n\left(\frac{1}{k+x}\right) \right\}. \quad (0 \leqq x \leqq 1, n \geqq 0) \tag{71}$$

In the event, by reason of the obvious relationship

$$z_n = \frac{1}{a_{n+1}+z_{n+1}},$$

the inequality

$$z_{n+1} < x$$

is satisfied if and only if for a suitably chosen positive integer k, the inequalities

$$\frac{1}{k+x} < z_n \leqq \frac{1}{k}$$

are fulfilled. Since, however, the measure of the set of numbers which satisfy these inequalities is clearly equal to

$$m_n\left(\frac{1}{k}\right) - m_n\left(\frac{1}{k+x}\right),$$

the relationship (71) follows.

It is easy to verify that for any C the function

$$\phi(x) = C \ln (1+x)$$

satisfies the relationship

$$\phi(x) = \sum_{k=1}^{\infty} \phi\left(\frac{1}{k}\right) - \phi\left(\frac{1}{k+x}\right).$$

This fact, apparently, was the pointer which led Gauss to the discovery of the correct expression for the limit of the function $m_n(x)$ as $n \to \infty$.

Formal differentiation of equation (72) yields

$$m'_{n+1}(x) = \sum_{k=1}^{\infty} \frac{1}{(k+x)^2} m'_n\left(\frac{1}{k+x}\right). \tag{72}$$

It is easy to show that equation (72) is rigorously true: since from $z_0(\alpha) = \alpha$ we have $m_0(x) = x$ and consequently $m'_0(x) = 1$. If for any n the function $m'_n(x)$ is bounded and continuous, then the series

standing on the right hand side of (72) is uniformly convergent. Hence, from a known theorem on the term by term differentiation of series, the sum of this series is also bounded, continuous, and equal to $m'_{n+1}(x)$. The relationship (72) is thus proved by induction.

For the purpose of our inquiry, equation (72) is considerably more convenient than equation (71). The principal result of Kusmin (to whose proof we now proceed) turns upon it.

THEOREM 33. *Let* $f_1(x), f_2(x), \ldots, f_n(x), \ldots$ *be a sequence of real functions defined in the interval* $(0, 1)$ *and there satisfying the relationship*

$$f_{n+1}(x) = \sum_{k=1}^{\infty} \frac{1}{(k+x)^2} f_n\left(\frac{1}{k+x}\right). \qquad n \geqq 0 \tag{73}$$

If, for $0 \leqq x \leqq 1$,

$$0 < f_0(x) < M \text{ and } |f_0(x)| < \mu,$$

then

$$f_n(x) = \frac{a}{1+x} + \theta A e^{-\lambda\sqrt{n}} \qquad (0 \leqq x \leqq 1)$$

with

$$a = \frac{1}{\ln 2} \int_0^1 f_0(z)dz \qquad |\theta| < 1.$$

Here λ *is an absolute positive constant and* A *is a positive constant which depends solely on* M *and* μ.

Since the proof is rather involved, we shall first establish some elementary lemmas.

LEMMA 1. *For any* $n \geqq 0$

$$f_n(x) = \sum^{(n)} f_0\left(\frac{p_n+xp_{n-1}}{q_n+xq_{n-1}}\right) \frac{1}{(q_n+xq_{n-1})^2} \tag{74}$$

Here $\left(\frac{p_n}{q_n}, \frac{p_n+p_{n-1}}{q_n+q_{n-1}}\right)$ *is an arbitrary interval of rank* n; *the summation is extended over all intervals of rank* n (*or, equivalently, over the elements* $a_1, a_2, \ldots, a_n$ *from* 1 *to* ∞).

PROOF. When $n = 0$ equation (74) is trivial, since in this case only one interval $(0, 1)$ is concerned, whence $p_0 = 0$, $q_0 = 1$, $p_{-1} = 1$ and $q_{-1} = 0$. Now let us assume that the relationship (74) is satisfied for a certain n. Then from the fundamental equation (73)

$$f_{n+1}(x) = \sum_{k=1}^{\infty} \frac{1}{(k+x)^2} f_n\left(\frac{1}{k+x}\right)$$

$$= \sum_{k=1}^{\infty} \frac{1}{(k+x)^2} \sum^{(n)} f_0\left(\frac{p_n + \dfrac{1}{k+x} p_{n-1}}{q_n + \dfrac{1}{k+x} q_{n-1}}\right) \frac{1}{\left(q_n + \dfrac{1}{k+x} q_{n-1}\right)^2}$$

$$= \sum^{(n)} \sum_{k=1}^{\infty} f_0\left\{\frac{(p_n k + p_{n-1}) + x p_n}{(q_n k + q_{n-1}) + x q_n}\right\} \frac{1}{\{(q_n k + q_{n-1}) + x q_n\}^2}$$

$$= \sum^{(n+1)} f_0\left(\frac{p_{n+1} + x p_n}{q_{n+1} + x q_n}\right) \frac{1}{(q_{n+1} + x q_n)^2}$$

which was to be proved.

LEMMA 2. *Under the assumptions of Theorem* 33

$$|f_n'(x)| < \frac{\mu}{2^{n-3}} + 4M.$$

PROOF. Differentiating equation (74) term by term, we obtain

$$f_n'(x) = \sum^{(n)} f_0'(u) \frac{(-1)^{n-1}}{(q_n + x q_{n-1})^4} - 2 \sum^{(n)} f_0(u) \frac{q_{n-1}}{(q_n + x q_{n-1})^3}.$$

Here we have written

$$u = \frac{p_n + x p_{n-1}}{q_n + x q_{n-1}}.$$

The term by term differentiation is justified by the uniform convergence of both sums on the right hand side, when $0 \leqq x \leqq 1$. Bearing in mind that

$$\frac{1}{(q_n + x q_{n-1})^2} < \frac{2}{q_n (q_n + q_{n-1})}$$

and taking note of the fact that from Theorem 12 (Chapter I)

$$q_n (q_n + q_{n-1}) > q_n^2 > 2^{n-1},$$

and finally taking the obvious relationship

$$\sum^{(n)} \frac{1}{q_n(q_n+q_{n-1})} = \sum^{(n)} \left| \frac{p_n}{q_n} - \frac{p_n+p_{n-1}}{q_n+q_{n-1}} \right| = 1$$

into account, then we obtain from the conditions of Theorem 33

$$|f_n'(x)| < \frac{\mu}{2^{n+3}} + 4M$$

which was to be proved.

LEMMA 3. *If*

$$\frac{t}{1+x} < f_n(x) < \frac{T}{1+x} \qquad (0 \leqq x \leqq 1)$$

then also

$$\frac{t}{1+x} < f_{n+1}(x) < \frac{T}{1+x}. \qquad (0 \leqq x \leqq 1)$$

PROOF. Under the conditions of the lemma the fundamental relationships (73) yield

$$\sum_{k=1}^{\infty} \frac{t}{1+\dfrac{1}{k+x}} \frac{1}{(k+x)^2} < f_{n+1}(x) < \sum_{k=1}^{\infty} \frac{T}{1+\dfrac{1}{k+x}} \frac{1}{(k+x)^2}$$

or

$$t \sum_{k=1}^{\infty} \frac{1}{(k+x)(k+x+1)} < f_{n+1}(x) < T \sum_{k=1}^{\infty} \frac{1}{(k+x)(k+x+1)}$$

or

$$t \sum_{k=1}^{\infty} \left(\frac{1}{k+x} - \frac{1}{k+x+1} \right) < f_{n+1}(x) < T \sum_{k=1}^{\infty} \left(\frac{1}{k+x} - \frac{1}{k+x+1} \right)$$

or finally

$$\frac{t}{1+x} < f_{n+1}(x) < \frac{T}{1+x},$$

which was to be proved.

LEMMA 4.

$$\int_0^1 f_n(z)dz = \int_0^1 f_0(z)dz \qquad (n = 0, 1, \ldots).$$

PROOF. From the fundamental equation (73)

$$\int_0^1 f_n(z)dz = \sum_{k=1}^{\infty} \int_0^1 f_{n-1}\left(\frac{1}{k+z}\right) \frac{dz}{(k+z)^2}$$

$$= \sum_{k=1}^{\infty} \int_{1/k+1}^{1/k} f_{n-1}(u)du = \int_0^1 f_{n-1}(u)du,$$

for $n > 0$, whence the assertion of Lemma 4 follows by induction.

PROOF OF THEOREM 33. By assumption the function $f_0(x)$ is differentiable and therefore continuous for $0 \leqq x \leqq 1$. Moreover since it was assumed to be positive in this interval then it possesses there a positive minimum which we shall denote by m. From the condition $m \leqq f_0(x) < M$ $(0 \leqq x \leqq 1)$ we clearly have

$$\frac{m}{2(1+x)} < f_0(x) < \frac{2M}{1+x} \qquad (0 \leqq x \leqq 1)$$

or

$$\frac{g}{1+x} < f_0(x) < \frac{G}{1+x}, \qquad (0 \leqq x \leqq 1)$$

where

$$g = \frac{m}{2}, \quad G = 2M.$$

We put in turn

$$f_n(x) - \frac{g}{1+x} = \phi_n(x). \qquad (0 \leqq x \leqq 1,\ n = 0,\ 1,\ 2,\ \ldots)$$

From Lemma 3 the function $F(x) = \dfrac{g}{1+x}$ satisfies the equation

$$F(x) = \sum_{k=1}^{\infty} F\left(\frac{1}{k+x}\right) \frac{1}{(k+x)^2}$$

(as can easily be verified independently). From this it clearly follows that the sequence of functions

$$\phi_0(x),\ \phi_1(x),\ \ldots,\ \phi_n(x),\ \ldots$$

satisfies equation (73). Thus all conclusions which we have drawn for the solution to this equation relate in equal measure to this sequence. In particular equation (74) holds. Thus, as before, writing

$$u = \frac{p_n + xp_{n-1}}{q_n + xq_{n-1}}$$

for conciseness, we obtain

$$\phi_n(x) = \sum^{(n)} \phi_0(u) \frac{1}{(q_n+xq_{n-1})^2}.$$

From this, taking into account the obvious inequalities

$$q_n+xq_{n-1} \leqq q_n+q_{n-1} < 2q_n \text{ and } \phi_0(u) > 0,$$

we finally derive

$$\phi_n(x) > \frac{1}{2} \sum^{(n)} \phi_0(u) \frac{1}{q_n(q_n+q_{n-1})}. \tag{75}$$

On the other hand, the mean value theorem yields

$$\frac{1}{2}\int_0^1 \phi_0(z)dz = \frac{1}{2} \sum^{(n)} \phi_0(u') \frac{1}{q_n(q_n+q_{n-1})}, \tag{76}$$

where u' is some point in the interval $\left(\frac{p_n}{q_n}, \frac{p_n+p_{n-1}}{q_n+q_{n-1}}\right)$ and $\frac{1}{q_n(q_n+q_{n-1})}$ is the length of this interval. Relationships (75) and (76) yield

$$\phi_n(x) - \frac{1}{2}\int_0^1 \phi_0(z)dz > \frac{1}{2} \sum^{(n)} \{\phi_0(u)-\phi_0(u')\} \frac{1}{q_n(q_n+q_{n-1})}. \tag{77}$$

Since, quite clearly,

$$|\phi_0(x)| \leqq |f_0'(x)|+g < \mu+g, \qquad (0 \leqq x \leqq 1)$$

we have

$$|\phi_0(u)-\phi_0(u')| < (\mu+g)|u-u'| < \frac{\mu+g}{q_n(q_n+q_{n-1})}$$
$$< \frac{\mu+g}{q_n^2} < \frac{\mu+g}{2^{n-1}},$$

and thus, from the inequality (77)

$$\phi_n(z) > \frac{1}{2}\int_0^1 \phi_0(z)dz - \frac{\mu+g}{2^n} = l - \frac{\mu+g}{2^n}$$

where

$$l = \frac{1}{2}\int_0^1 \phi_0(z)dz.$$

Thus we obtain

$$f_n(x) > \frac{g}{1+x} + l - \frac{\mu+g}{2^n} > \frac{g+l-2^{-n+1}(\mu+g)}{(1+x)} = \frac{g_1}{1+x}.$$

In a completely analogous way, considering the sequence of functions

$$\psi_n(x) = \frac{G}{1+x} - f_n(x) \quad (n = 0, 1, 2, \ldots)$$

and reasoning as before, we obtain the inequality

$$f_n(x) < \frac{G-l'+2^{-n+1}(\mu+g)}{1+x} = \frac{G_1}{1+x}$$

where

$$l' = \frac{1}{2}\int_0^1 \psi_0(z)dz.$$

Since $l > 0$ and $l' > 0$, then for sufficiently large n

$$g < g_1 < G_1 < G$$

and

$$G_1 - g_1 < G - g - (l+l') + 2^{-n+2}(\mu+G).$$

From

$$l+l' = \frac{1}{2}\int_0^1 \frac{G-g}{1+z}dz = (G-g)\frac{\ln 2}{2}$$

however,

$$G_1 - g_1 < (G-g)\delta + 2^{-n+2}(\mu+G)$$

where

$$\delta = 1 - \frac{\ln 2}{2} < 1$$

is an absolute positive constant.

Let us at this stage assemble the results which we have derived: *From the conditions*

$$\frac{g}{1+x} < f_0(x) < \frac{G}{1+x}, \quad |f_0'(x)| < \mu$$

we obtain for sufficiently large n

$$\frac{g_1}{1+x} < f_n(x) < \frac{G_1}{1+x},$$

where $g < g_1 < G_1 < G$ and $G_1 - g_1 < (G-g)\delta + 2^{-n+2}(\mu+G)$.

Now taking as the starting function not $f_0(x)$, but $f_n(x)$, and repeating the previous reasoning, we evidently arrive at the inequality

$$\frac{g_2}{1+x} < f_{2n}(x) < \frac{G_2}{1+x},$$

where

$$\begin{cases} g_1 < g_2 < G_2 < G_1, \\ G_2 - g_2 < \delta(G_1 - g_1) + 2^{-n+2}(\mu_1 + G_1) \end{cases}$$

and μ_1 is a positive integer for which

$$|f_n'(x)| < \mu_1. \qquad (0 \leqq x \leqq 1)$$

Continuing this process then we obtain quite generally the relationship

$$\frac{g_r}{1+x} < f_{rn}(x) < \frac{G_r}{1+x} \qquad (0 \leqq x \leqq 1; r = 0, 1, 2, \ldots)$$

where, for $r > 0$,

$$\begin{cases} g_{r-1} < g_r < G_r < G_{r-1} \\ G_r - g_r < \delta(G_{r-1} - g_{r-1}) + 2^{-n+2}(\mu_{r-1} + G_{r-1}). \end{cases} \tag{78}$$

Here μ_{r-1} is a positive number satisfying the condition

$$|f_{(r-1)n}'(x)| < \mu_{r-1}. \qquad (0 \leqq x \leqq 1)$$

From Lemma 2, we can put

$$\mu_r = \frac{\mu}{2^{rn-3}} + 4M \qquad (r = 0, 1, 2, \ldots)$$

and consequently, choosing n to be sufficiently large,

$$\mu_r < 5M. \quad (r = 1, 2, \ldots)$$

Hence, successive application of inequality (78) yields for $r = 1, 2, \ldots, n$

$$G_n - g_n < (G-g)\delta^n + 2^{-n+2}\{(\mu+2M)\delta^{n-1} + 7M\delta^{n-2} + 7M\delta^{n-3} + \ldots + 7M\delta + 7M\}.$$

Since $\delta < 1$ is an absolute constant, it follows that

$$G_n - g_n < Be^{-\lambda n}$$

where λ is an absolute constant, whilst $B > 0$ depends only on M and μ.

From this it follows that a common limit

$$\lim_{n\to\infty} G_n = \lim_{n\to\infty} g_n = a$$

exists, and that

$$\left| f_{n^2}(x) - \frac{a}{1+x} \right| < Be^{-\lambda n}, \quad (0 \leqq x \leqq 1) \tag{79}$$

from which, in particular

$$\int_0^1 f_{n^2}(z)dz \to a \ln 2, \qquad (n \to \infty)$$

and hence, from Lemma 4, there follows

$$a = \frac{1}{\ln 2} \int_0^1 f_0(z)dz.$$

Finally let $n^2 \leqq N < (n+1)^2$. Since, from the inequality (79)

$$\frac{a - 2Be^{-\lambda n}}{1+x} < f_{n^2}(x) < \frac{a + 2Be^{-\lambda n}}{1+x}$$

there follows from Lemma 3

$$\frac{a - 2Be^{-\lambda n}}{1+x} < f_N(x) < \frac{a + 2Be^{-\lambda n}}{1+x}$$

or

$$\left| f_N(x) - \frac{a}{1+x} \right| < 2Be^{-\lambda n} = Ae^{-\lambda(n+1)} < Ae^{-\lambda\sqrt{N}},$$

in which we have put $A = 2Be^{\lambda}$. This inequality, which has been derived for sufficiently large N, can clearly be made to hold for all $N \geqq 0$ by making A suitably large. Hence Theorem 33 has been proved in its entirety.

Now let us return to Gauss' problem and put

$$f_n(x) = m_n'(x). \qquad (0 \leqq x \leqq 1)$$

Then $f_0(x) \equiv 1$; hence all conditions of Theorem 33 are fulfilled, and we obtain

$$\left| m_n'(x) - \frac{1}{(1+x)\ln 2} \right| < A e^{-\lambda\sqrt{n}}; \quad (0 \leqq x \leqq 1) \tag{80}$$

integrating this, there follows

$$\left| m_n(x) - \frac{\ln(1+x)}{\ln 2} \right| < A e^{-\lambda\sqrt{n}}. \qquad (0 \leqq x \leqq 1)$$

Here A and λ are absolute positive constants. In this way not only has Gauss' conjecture been proved, but also a good estimate for the remainder term has been obtained[1]).

We shall now apply this result to estimate the measure of the set of numbers for which $a_n = k$ for large n.

Since the condition $a_n = k$ is clearly equivalent to the inequality

$$\frac{1}{k+1} < z_{n-1} \leqq \frac{1}{k},$$

we have

$$\mathfrak{M}E\binom{n}{k} = m_{n-1}\left(\frac{1}{k}\right) - m_{n-1}\left(\frac{1}{k+1}\right) = \int_{1/(k+1)}^{1/k} m_{n-1}'(x)\,dx,$$

from which, on account of the inequality (80)

$$\left| \mathfrak{M}E\binom{n}{k} - \frac{\ln\left\{1+\frac{1}{k(k+2)}\right\}}{\ln 2} \right| < \frac{A}{k(k+1)} e^{-\lambda\sqrt{n-1}}. \tag{81}$$

From this inequality we now obtain for the quantity $\mathfrak{M}E\binom{n}{k}$ (for which we obtained an extremely crude estimate in § 13) the exact limiting value relationship

$$\mathfrak{M}E\binom{n}{k} \to \frac{\ln\left\{1+\frac{1}{k(k+2)}\right\}}{\ln 2}. \qquad (n \to \infty).$$

---

[1]) The method of P. Lévy allows a better estimate to be obtained. The inequality

$$\left| m_n(x) - \frac{\ln(1+x)}{\ln 2} \right| < \mathrm{A}e^{-\lambda n} \qquad (0 \leqq x \leqq 1)$$

is shown to be satisfied. (B.G.)

Thus, for example, the measure of the set of numbers for which $a_n = 1$, tends, as $n \to \infty$, to the quantity

$$\frac{\ln 4 - \ln 3}{\ln 2}.$$

Apart from the proof of the Gauss conjecture, Theorem 33 enables us to derive a further, more general and extremely important result of which we shall make use in the future. We denote by $M_n(x)$ the measure of the set of numbers belonging to a fixed interval of rank k, and moreover for which the condition $z_{k-n} < x$ is satisfied. In other words $M_n(x)$ is the measure of the set of numbers in the interval $(0, 1)$ for which the conditions

$$a_1 = r_1, a_2 = r_2, \ldots, a_k = r_k; z_{k+n} < x \tag{82}$$

are satisfied, where the $r_1, r_2, \ldots, r_n$ are certain fixed positive integers, whilst $n \geqq 0$ and x $(0 \leqq x \leqq 1)$ can vary arbitrarily.

If conditions (82) are to be fulfilled then it is evidently necessary and sufficient that the conditions

$$a_1 = r_1, a_2 = r_2, \ldots, a_k = r_k; \frac{1}{r+x} < z_{k+n-1} \leqq \frac{1}{r}$$

should hold; here r is a positive integer. From this follows

$$M_n(x) = \sum_{r=1}^{\infty} \left\{ M_{n-1}\left(\frac{1}{r}\right) - M_{n-1}\left(\frac{1}{r+x}\right) \right\}, \quad (n \geqq 1, 0 \leqq x \leqq 1)$$

whence the sequence of functions $M_0'(x), M_1'(x), \ldots, M_n'(x)$ satisfies equation (73).

An arbitrary number α in the interval $\left(\frac{p_k}{q_k}, \frac{p_k+p_{k-1}}{q_k+q_{k-1}}\right)$ can be expressed in the form

$$\alpha = \frac{p_k r_{k+1} + p_{k-1}}{q_k r_{k+1} + q_{k-1}}$$

or since $z_k = 1/r_{k+1}$, in the form,

$$\alpha = \frac{p_k + z_k p_{k-1}}{q_k + z_k q_{k-1}}.$$

For $z_k < x$ the number α must be enclosed between $\frac{p_k}{q_k}$ and $\frac{p_k + x p_{k-1}}{q_k + x q_{k-1}}$, therefore

$$M_0(x) = \left| \frac{p_k}{q_k} - \frac{p_k + x p_{k-1}}{q_k + x q_{k-1}} \right| = \frac{x}{q_k(q_k + x q_{k-1})}. \tag{83}$$

Now putting

$$M_n(x) = \mathfrak{M}E \begin{pmatrix} 1, & 2, & \ldots, & k \\ r_1, & r_2, & \ldots, & r_k \end{pmatrix} \chi_n(x), \quad (n \geqq 0,\ 0 \leqq x \leqq 1)$$

we obtain a new sequence of functions

$$\chi_0(x),\ \chi_1(x),\ \ldots,\ \chi_n(x).$$

Here the functions $\chi_n'(x)$, which differ from the corresponding functions $M_n'(x)$ only by a constant factor, also satisfy equation (73). Since, clearly,

$$\mathfrak{M}E \begin{pmatrix} 1, & 2, & \ldots, & k \\ r_1, & r_2, & \ldots, & r_k \end{pmatrix} = \left| \frac{p_k}{q_k} - \frac{p_k + p_{k-1}}{q_k + q_{k-1}} \right| = \frac{1}{q_k(q_k + q_{k-1})},$$

equation (83) yields

$$\chi_0(x) = \frac{(q_k + q_{k-1})x}{q_k + q_{k-1}x},$$

from which

$$\chi_0'(x) = \frac{q_k(q_k + q_{k-1})}{(q_k + q_{k-1}x)^2}$$

and

$$\chi_0''(x) = -\frac{2q_k q_{k-1}(q_k + q_{k-1})}{(q_k + q_{k-1}x)^3}.$$

Thus

$$\frac{1}{2} < \chi_0'(x) < 2, \qquad |\chi_0''(x)| < 4 \qquad (0 \leqq x \leqq 1).$$

These show that Theorem 23 can be applied to the sequence of functions $\chi_n'(x)$. Thus it transpires that the numbers A and k are absolute constants, i.e. they do not depend on the $r_1, r_2, \ldots, r_n$; we obtain

$$\chi_n'(x) = \frac{M_n'(x)}{\mathfrak{M}E \begin{pmatrix} 1, & 2, & \ldots, & k \\ r_1, & r_2, & \ldots, & r_k \end{pmatrix}} = \frac{1}{(1+x)\ln 2} + \theta e^{-\lambda\sqrt{n}},\ |\theta| < 1.$$

Integrating this relationship between the limits $1/(r+1)$ and $1/r$ (r is a positive integer) we obtain for $|\theta'| < 1$

$$\frac{M_n\left(\frac{1}{r}\right)-M_n\left(\frac{1}{r+1}\right)}{\mathfrak{M}E\begin{pmatrix}1,2,\ \ldots,k\\ r_1,r_2,\ \ldots,r_k\end{pmatrix}}=\frac{\ln\left\{1+\frac{1}{r(r+2)}\right\}}{\ln 2}+\frac{\theta' A}{r(r+1)}e^{-\lambda\sqrt{n}}.$$

Since, clearly,

$$M_n\left(\frac{1}{r}\right)-M_n\left(\frac{1}{r+1}\right)=\mathfrak{M}E\begin{pmatrix}1,\ 2,\ \ldots,\ k,\ k+n-1\\ r_1,\ r_2,\ \ldots,\ r_k,\ \ \ r\end{pmatrix},$$

there follows

$$\mathfrak{M}E\begin{pmatrix}1,\ 2,\ \ldots,\ k,\ k+n-1\\ r_1,\ r_2,\ \ldots,\ r_k,\ \ \ r\end{pmatrix}$$
$$=\left(\frac{\ln\left\{1+\frac{1}{r(r+2)}\right\}}{\ln 2}+\frac{\theta' A e^{-\lambda\sqrt{n}}}{r(r+1)}\right)ME\begin{pmatrix}1,\ 2,\ \ldots,\ k\\ r_1,\ r_2,\ \ldots,\ r_k\end{pmatrix}.$$

Finally we can sum this relationship between the limits 1 and ∞ over certain (arbitrarily chosen) of the numbers $r_1, r_2, \ldots, r_n$. By doing this the corresponding indices simply vanish from both states of the relationship; in this way we obtain in place of the successive indices a series of completely arbitrary indices $n_1, n_2, \ldots, n_t$; the relationship retains its validity completely. In this way we obtain

THEOREM 34. *There exist two absolute positive constants A and λ such that for $n_1 < n_2 < \ldots < n_t < n_{t+1}$ and for any positive integers $r_1, r_2, \ldots, r_t, r$ the inequality*

$$\left|\frac{\mathfrak{M}E\begin{pmatrix}n_1,\ n_2,\ \ldots,\ n_t,\ n_{t+1}\\ r_1,\ r_2,\ \ldots,\ r_t,\ \ r\end{pmatrix}}{\mathfrak{M}E\begin{pmatrix}n_1,\ n_2,\ \ldots,\ n_t\\ r_1,\ r_2,\ \ldots,\ r_t\end{pmatrix}}-\frac{\ln\left\{1+\frac{1}{r(r+2)}\right\}}{\ln 2}\right|$$
$$<\frac{A}{r(r+1)}e^{-\lambda\sqrt{n_{t+1}-n_t-1}}$$

holds.

This result shows that not only does the set of points in the interval (0, 1) for which $a_n = r$ have a measure which tends to a fixed limit as $n \to \infty$, but that the set of points which satisfy this condition and whose previous elements are fixed have a relative measure which also has the same limit.

§ 16 Mean Values [1])

The results of the previous section provide us with the possibility of proving the following general theorem.

THEOREM 35. *Let $f(r)$ be a non-negative function of the positive integer r. Further let positive constants C and δ exist such that*

$$f(r) < Cr^{\frac{1}{2}-\delta}. \qquad (r = 1, 2, \ldots)$$

Then for all numbers α in the interval $(0, 1)$, with the exception of those of a set of measure zero

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} f(a_k) = \sum_{r=1}^{\infty} f(r) \frac{\ln\left\{1+\frac{1}{r(r+2)}\right\}}{\ln 2} \qquad (84)$$

holds.

PRELIMINARY REMARK. The convergence of the series on the right hand side of equation (84) clearly follows from the conditions which have been imposed on $f(r)$.

PROOF. We put

$$\int_0^1 f(a_k)d\alpha = u_k, \quad \int_0^1 \{f(a_k)-u_k\}^2 d\alpha = b_k,$$

$$\int_0^1 \{f(a_i)-u_i\}\{f(a_k)-u_k\}d\alpha = g_{ik},$$

$$\sum_{k=1}^{n} \{f(a_k)-u_k\} = s_n = s_n(\alpha).$$

The existence of all the integrals mentioned is a simple consequence of the conditions imposed upon $f(r)$. Since

$$\{f(r)\}^2 < C^2 r^{1-2\delta}$$

the integral

$$\int_0^1 \{f(a_k)\}^2 d\alpha = \sum_{r=1}^{\infty} \{f(r)\}^2 \mathfrak{M}E\binom{k}{r} < 2C^2 \sum_{r=1}^{\infty} \frac{r^{1-2\delta}}{r^2} = C_1$$

is indeed meaningful, so that from the Schwarz-Bunjakowski inequality the existence of the mentioned integrals follows. In particular there follow

[1]) The results of this section are given in the article by A. Ya. Khintchine, "Metrische Kettenbruchprobleme", Compositio Mathematica, vol. 1, 1935, 361-382. (B.G.)

$$\left.\begin{aligned} b_k &= \int_0^1 \{f(a_k)\}^2 d\alpha - u_k^2 < C_1, \\ u_k &= \int_0^1 f(a_k) d\alpha < \sqrt{\int_0^1 \{f(a_k)\}^2 d\alpha} < \sqrt{C_1}. \end{aligned}\right\} \tag{85}$$

Furthermore for $k > i$ we clearly have

$$\begin{aligned} g_{ik} &= \int_0^1 f(a_i) f(a_k) d\alpha - u_i u_k \\ &= \sum_{r,s=1}^{\infty} f(r) f(s) \mathfrak{M} E \binom{i, k}{r, s} - u_i u_k. \end{aligned} \tag{86}$$

From Theorem 34 and the concluding inequality of § 12, however,

$$\begin{aligned} &\left| \mathfrak{M} E \binom{i, k}{r, s} - \frac{\ln\left\{1 + \frac{1}{s(s+2)}\right\}}{\ln 2} \mathfrak{M} E \binom{i}{r} \right| \\ &\quad < \frac{A e^{-\lambda\sqrt{k-i-1}}}{s(s+1)} \mathfrak{M} E \binom{i}{r} \\ &\quad < 3A e^{-\lambda\sqrt{k-i-1}} \mathfrak{M} E \binom{i}{r} \mathfrak{M} E \binom{k}{s} \end{aligned} \tag{87}$$

and

$$\begin{aligned} &\left| \mathfrak{M} E \binom{k}{s} - \frac{\ln\left\{1 + \frac{1}{s(s+2)}\right\}}{\ln 2} \right| \\ &\quad < \frac{A e^{-\lambda\sqrt{k-1}}}{s(s+1)} < 3A e^{-\lambda\sqrt{k-1}} \mathfrak{M} E \binom{k}{s}. \end{aligned} \tag{88}$$

Now multiplying the inequality (88) throughout by $\mathfrak{M} E \binom{i}{r}$ and comparing the result obtained with inequality (87), we obtain

$$\begin{aligned} &\left| \mathfrak{M} E \binom{i, k}{r, s} - \mathfrak{M} E \binom{i}{r} \mathfrak{M} E \binom{k}{s} \right| \\ &\quad < 6A e^{-\lambda\sqrt{k-i-1}} \mathfrak{M} E \binom{i}{r} \mathfrak{M} E \binom{k}{s}, \end{aligned}$$

whence the relationship (86) yields the inequality

$$\left| g_{ik} - \sum_{r,s=1}^{\infty} f(r)f(s)\mathfrak{M}E\binom{i}{r}\mathfrak{M}E\binom{k}{s} + u_i u_k \right|$$
$$< 6Ae^{-\lambda\sqrt{k-i-1}} \sum_{r,s=1}^{\infty} f(r)f(s)\mathfrak{M}E\binom{i}{r}\mathfrak{M}E\binom{k}{s}.$$

Noting that

$$\sum_{r,s=1}^{\infty} f(r)f(s)\,\mathfrak{M}E\binom{i}{r}\mathfrak{M}E\binom{k}{s} = u_i u_k$$

and using the second of inequalities (85) to estimate the right hand side, we obtain

$$|g_{ik}| < 6Ae^{-\lambda\sqrt{k-i-1}}u_i u_k < 6AC_1 e^{-\lambda\sqrt{k-i-1}}. \tag{89}$$

Applying the estimates (85) and (89) we find that for $n > m > 0$

$$\int_0^1 (s_n - s_m)^2 d\alpha = \int_0^1 \left\{ \sum_{k=m+1}^{n} (f(a_k) - u_k) \right\}^2 d\alpha$$
$$= \sum_{k=m+1}^{n} \int_0^1 \{f(a_k) - u_k\}^2 d\alpha$$
$$+ 2 \sum_{i=m+1}^{n} \sum_{k=i+1}^{n} \int_0^1 \{f(a_i) - u_i\}\{f(a_k) - u_k\} d\alpha$$
$$= \sum_{k=m+1}^{n} b_k + 2 \sum_{i=m+1}^{n} \sum_{k=i+1}^{n} g_{ik}$$
$$< C_1(n-m) + 12AC_1 \sum_{i=m+1}^{n} \sum_{k=i+1}^{\infty} e^{-\lambda\sqrt{k-i-1}} < C_2(n-m), \tag{90}$$

where C_2 is a new positive constant.

Now we denote by e_n the set of points in the interval $(0, 1)$ for which

$$|s_n| \geqq \varepsilon n,$$

where ε is an arbitrarily prescribed positive number. Clearly

$$\int_0^1 s_n^2 d\alpha \geqq \int_{e_n} s_n^2 d\alpha \geqq \varepsilon^2 n^2 \mathfrak{M}e_n,$$

whence the inequality (90) (for $m = 0$) yields

$$\mathfrak{M}e_n \leqq \frac{\int_0^1 s_n^2 d\alpha}{\varepsilon^2 n^2} < \frac{C_2}{\varepsilon^2 n}.$$

Thus the series

$$\sum_{n=1}^{\infty} \mathfrak{M}e_{n^2}$$

converges and in consequence, as we know, almost all of the numbers in the interval (0, 1) belong to at most a finite number of the sets e_{n^2} $(n = 1, 2, 3, \ldots)$. This means that for almost all numbers in the interval (0, 1) the estimate

$$\frac{s_{n^2}}{n^2} < \varepsilon$$

holds for sufficiently large n. Since however, ε is arbitrarily small

$$\lim_{n\to\infty} \frac{s_{n^2}}{n^2} = 0. \tag{91}$$

Furthermore formula (90) yields for $n^2 \leqq N \leqq (n+1)^2$

$$\int_0^1 (s_N - s_{n^2}) d\alpha < C_2(N-n^2) < C_2(2n+1) \leqq 3C_2 n.$$

Denoting by $e_{n,N}$ the set of points in the interval (0, 1) for which $|s_N - s_{n^2}| \geqq \varepsilon n^2$, and putting

$$\sum_{N=n^2}^{(n+1)^2-1} e_{n,N} = E_n,$$

then we obtain for $n^2 \leqq N < (n+1)^2$

$$\int_0^1 (s_N - s_{n^2})^2 d\alpha \geqq \int_{e_{n,N}} (s_N - s_{n^2})^2 d\alpha > \varepsilon^2 n^4 \mathfrak{M}e_{n,N},$$

$$\mathfrak{M}e_{n,N} < \frac{3C_2}{\varepsilon^2 n^3},$$

$$\mathfrak{M}E_n \leqq \sum_{N=n^2}^{(n+1)^2-1} \mathfrak{M}e_{n,N} < \frac{3C_2(2n+1)}{\varepsilon^2 n^3} \leqq \frac{9C_2}{\varepsilon^2 n^2},$$

whence the series $\sum_{n=1}^{\infty} \mathfrak{M}E_n$ converges. Almost all numbers in the interval (0, 1) belong therefore, as we know, to at most finitely many of the sets E_n and hence of the sets $e_{n,N}$. This means, however, that for sufficiently large n and for $n^2 \leqq N \leqq (n+1)^2$ almost all numbers in the interval (0, 1) satisfy the inequality

$$|s_N - s_{n^2}| < \varepsilon n^2.$$

In other words, for sufficiently large n and for $n^2 \leqq N \leqq (n+1)^2$

$$\frac{|s_N - s_{n^2}|}{n^2} < \varepsilon$$

almost everywhere, and, since ε may be made arbitrarily small

$$\frac{s_N}{n^2} - \frac{s_{n^2}}{n^2} \to 0 \qquad [n \to \infty, n^2 \leqq N \leqq (n+1)^2].$$

From relationship (91) it clearly follows that almost everywhere

$$\frac{s_N}{n^2} \to 0, \qquad [n \to \infty, n^2 \leqq N < (n+1)^2]$$

and thus

$$\frac{s_N}{N} \to 0. \qquad (N \to \infty)$$

In other words

$$\frac{1}{N}\sum_{k=1}^{N} f(a_k) - \frac{1}{N}\sum_{k=1}^{N} u_k \to 0 \qquad (N \to \infty) \qquad (92)$$

holds almost everywhere. From formula (81) of the previous section, however,

$$\left| u_k - \sum_{r=1}^{\infty} f(r) \frac{\ln\left\{1+\frac{1}{r(r+2)}\right\}}{\ln 2} \right|$$

$$= \sum_{r=1}^{\infty} f(r) \left| \mathfrak{M}E\binom{k}{r} - \frac{\ln\left\{1+\frac{1}{r(r+2)}\right\}}{\ln 2} \right|$$

$$< Ae^{-\lambda\sqrt{k-1}} \sum_{r=1}^{\infty} \frac{f(r)}{r(r+1)} < A_1 e^{-\lambda\sqrt{k}},$$

where A_1 is a new positive constant. Thus

$$u_k \to \sum_{r=1}^{\infty} f(r) \frac{\ln\left\{1+\frac{1}{r(r+2)}\right\}}{\ln 2} \quad \text{for } k \to \infty$$

and consequently also

$$\frac{1}{N}\sum_{k=1}^{N} u_k \to \sum_{r=1}^{\infty} f(r) \frac{\ln\left\{1+\frac{1}{r(r+2)}\right\}}{\ln 2} \quad \text{for } N \to \infty.$$

Using this, it follows from relationship (92) that

$$\frac{1}{N}\sum_{k=1}^{N} f(a_k) \to \sum_{r=1}^{\infty} f(r)\,\frac{\ln\left\{1+\dfrac{1}{r(r+2)}\right\}}{\ln 2}$$

almost everywhere in (0, 1), whence Theorem 35 has been proved.

The theorem which has just been proved enables us to distinguish a series of properties of continued fractions which hold for almost all irrational numbers. We put, for example

$$f(r) = 1 \text{ for } r = k \text{ and } f(r) = 0 \text{ for } r \neq k$$

where k is an arbitrary positive integer. Clearly, in this case, the sum

$$\psi_n(k) = \sum_{i=1}^{n} f(a_i)$$

represents a number which tells us how many times the number k occurs in the first n elements of the continued fraction in question. The quotient

$$\frac{\psi_n(k)}{n} = \frac{1}{n}\sum_{i=1}^{n} f(a_i)$$

gives, in the same way, the *density of the number k in the first n elements of the continued fraction*. Finally the limit

$$\lim \frac{\psi_n(k)}{n} = d(k)$$

if indeed it exists, may naturally be interpreted as *the density of the number k in the sequence of elements of the continued fraction in question.*

Since the function $f(r)$ which we have defined clearly satisfies all the conditions of Theorem 35, we may conclude from this theorem that *for any k this density exists almost everywhere and remains the same almost everywhere*. Furthermore the same theorem enables us to compute the values of these densities: clearly we have almost everywhere

$$d(1) = \frac{\ln 4 - \ln 3}{\ln 2}, \; d(2) = \frac{\ln 9 - \ln 8}{\ln 2}, \; d(3) = \frac{\ln 16 - \ln 15}{\ln 2},$$

etc. Thus, in the mean, each positive integer occurs with the same frequency in the developments of almost all numbers.

We obtain a further interesting result if we put

$$f(r) = \ln r. \qquad (r = 1, 2, 3, \ldots)$$

Clearly all the conditions of Theorem 35 are fulfilled for this function. Therefore we find that

$$\frac{1}{n}\sum_{i=1}^{n} \ln a_i \to \sum_{r=1}^{\infty} \ln r \frac{\ln\left\{1+\frac{1}{r(r+2)}\right\}}{\ln 2} \qquad (n \to \infty)$$

holds almost everywhere, or equivalently

$$\sqrt[n]{a_1 a_2 \ldots a_n} \to \prod_{r=1}^{\infty} \left\{1+\frac{1}{r(r+2)}\right\}^{\frac{\ln r}{\ln 2}}.$$

Thus the geometric mean of the first n elements tends almost everywhere as $n \to \infty$ *to the absolute constant*

$$\prod_{r=1}^{\infty} \left\{1+\frac{1}{r(r+2)}\right\}^{\frac{\ln r}{\ln 2}} = 2 \cdot 6 \ldots.$$

Clearly Theorem 35 allows similar results for a whole series of other mean values to be derived. As far as the arithmetic mean

$$\frac{1}{n}\sum_{i=1}^{n} a_i \tag{93}$$

is concerned, however, examination by means of this method is not possible, since the corresponding function $f(r) = r$ does not satisfy the conditions of Theorem 35. Nevertheless it is quite clear from elementary considerations that the expression (93) possesses a finite limit almost nowhere. Indeed it follows from Theorem 30 (§ 13) that for infinitely many values of n

$$a_n > n \ln n$$

almost everywhere, and since

$$\sum_{i=1}^{n} a_i > n \ln n$$

it follows that

$$\frac{1}{n}\sum_{i=1}^{n} a_i > \ln n.$$

Thus the quantity (93) is almost everywhere unbounded and thus, as we asserted, does not possess a finite limit.

A. N. KHOVANSKII

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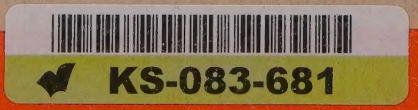
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